



DP 22

Analyst	beldon
Date	4/27/2026 11:29 AM

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System Information

Application	Fluent
Settings	3d, double precision, pressure-based, SST k-omega
Version	24.2.0-10192
Source Revision	149ed7492d
Build Time	May 13 2024 11:22:47 EST
CPU	AMD Ryzen 9 7950X 16-Core Processor
OS	Windows

Geometry and Mesh

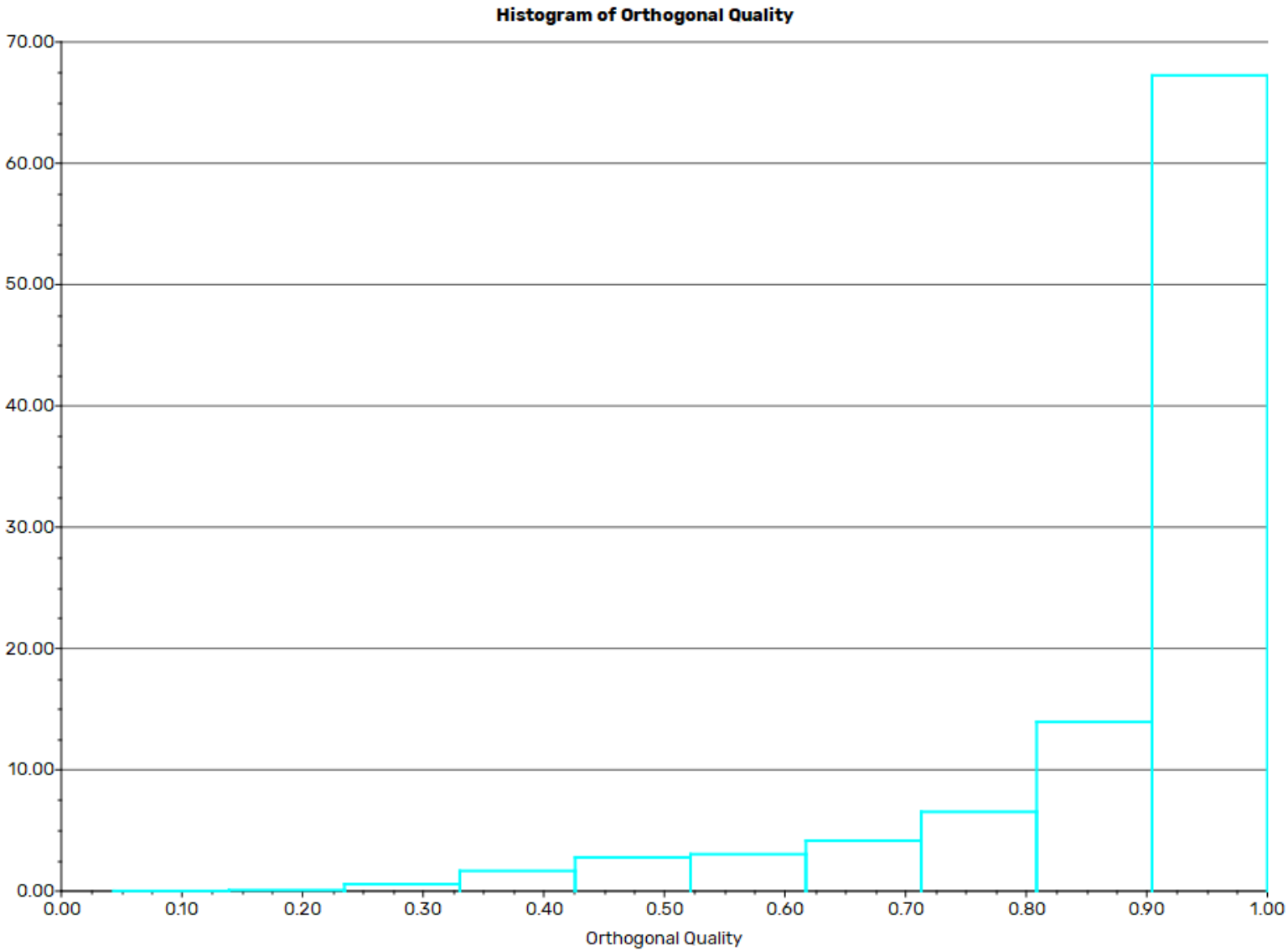
Mesh Size

Cells	Faces	Nodes
5304451	22302322	12232546

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
enclosure	Mixed Cell	0.043447666	1416.6115

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	SST k-omega turbulence model

Material Properties

— Fluid	
— air	
Density	1.225 kg/m^3
Cp (Specific Heat)	1006.43 J/(kg K)
Thermal Conductivity	0.0242 W/(m K)
Viscosity	1.7894e-05 kg/(m s)
Molecular Weight	28.966 kg/kmol
— Solid	
— aluminum	
Density	2719 kg/m^3
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)

Cell Zone Conditions

— Fluid	
— enclosure	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no
3D Fan Zone?	no

Boundary Conditions

— Inlet	
— inlet	
Velocity Specification Method	Magnitude, Normal to Boundary
Reference Frame	Absolute

Velocity Magnitude	inletv
Supersonic/Initial Gauge Pressure [Pa]	0
Turbulence Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
— Outlet	
— outlet	
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Build artificial walls to prevent reverse flow?	yes
Radial Equilibrium Pressure Distribution	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
— Symmetry	
symmetry	symmetry
— Wall	
— walls	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— boi_midplate_ib.1	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— rsus	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— fsus	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— fwing	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0

Wall Roughness Constant	0.5
— chasis	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— fwheel	
Wall Motion	Moving Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Define wall motion relative to adjacent cell zone?	yes
Apply a rotational velocity to this wall?	yes
Define wall velocity components?	no
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Rotation Speed	wheelspeed
X-Position of Rotation-Axis Origin [m]	0
Y-Position of Rotation-Axis Origin [m]	0
Z-Position of Rotation-Axis Origin [m]	0
— rwheel	
Wall Motion	Moving Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Define wall motion relative to adjacent cell zone?	yes
Apply a rotational velocity to this wall?	yes
Define wall velocity components?	no
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
Rotation Speed	wheelspeed
X-Position of Rotation-Axis Origin [m]	1.55
Y-Position of Rotation-Axis Origin [m]	0
Z-Position of Rotation-Axis Origin [m]	0
— rwing	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— undertray	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5

— ground	
Wall Motion	Moving Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Define wall motion relative to adjacent cell zone?	yes
Apply a rotational velocity to this wall?	no
Velocity Magnitude	inletv
X-Component of Wall Translation	1
Y-Component of Wall Translation	0
Z-Component of Wall Translation	0
Define wall velocity components?	no
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— boi_midplate_ib.1-shadow	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	20 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	enclosure

Solver Settings

— Equations	
Flow	True
Turbulence	True
— Numerics	
Absolute Velocity Formulation	True
— Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1

Turbulent Kinetic Energy	0.75
Specific Dissipation Rate	0.75
Turbulent Viscosity	1
Explicit Momentum	0.5
Explicit Pressure	0.5
— Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
— Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
— Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Spec. Dissipation Rate [s^-1]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	26
Case Read	37.551 seconds
Data Read	7.626 seconds
Virtual Current Memory	21.923 GB
Virtual Peak Memory	22.3791 GB
Memory Per M Cell	4.01518

Solution Status

Iterations: 200

	Value	Absolute Criteria	Convergence Status
continuity	0.01562611	0.001	Not Converged
x-velocity	1.705541e-06	0.001	Converged
y-velocity	1.049473e-06	0.001	Converged
z-velocity	8.943566e-07	0.001	Converged
k	0.001083133	0.001	Not Converged
omega	0.0007942878	0.001	Converged

Named Expressions

Expression	Definition	Value	Unit	Used In	Description
inletv	20	20	[m s^-1]	inlet (Velocity Magnitude) , ground (Velocity Magnitude) , wheelspeed (Named Expression)	
wheelspeed	inletv / 0.2032[m] * 1[rad]	98.425197	[s^-1 rad]	fwheel (Rotation Speed) , rwheel (Rotation Speed)	

Report Definitions

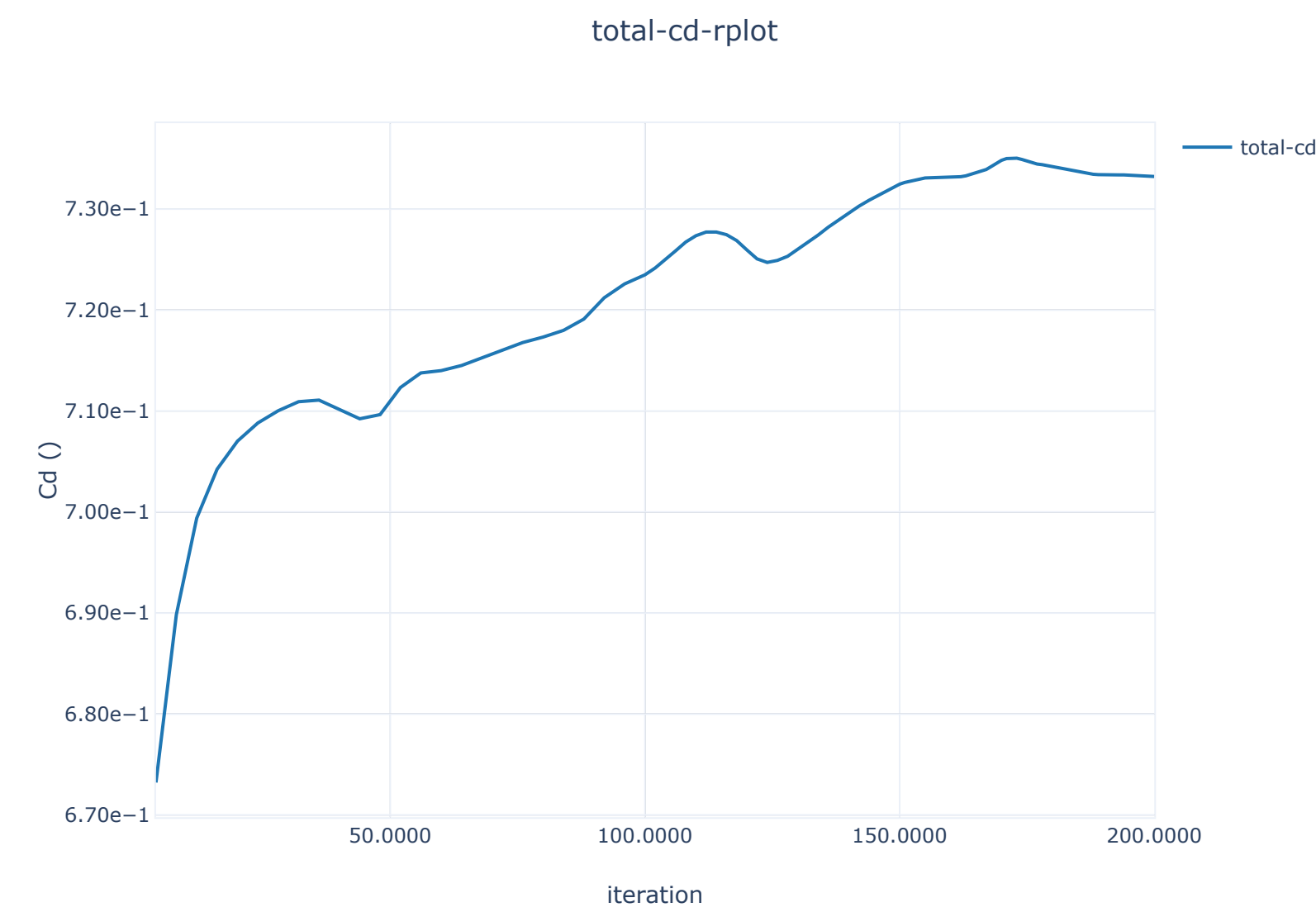
ut-cl	-0.7857225	
ut-lift	-192.502	N
ut-drag	25.31085	N
ut-cd	0.1033096	
fwing-cl	-0.6405706	
fwing-lift	-156.9398	N
fwing-drag	19.90826	N
fwing-cd	0.08125819	
rwing-cl	-0.5754741	
rwing-lift	-140.9912	N
rwing-drag	62.7296	N
rwing-cd	0.2560392	
total-lift	-486.6432	N
total-cl	-1.986299	
total-drag	179.6394	N
total-cd	0.733222	

Plots

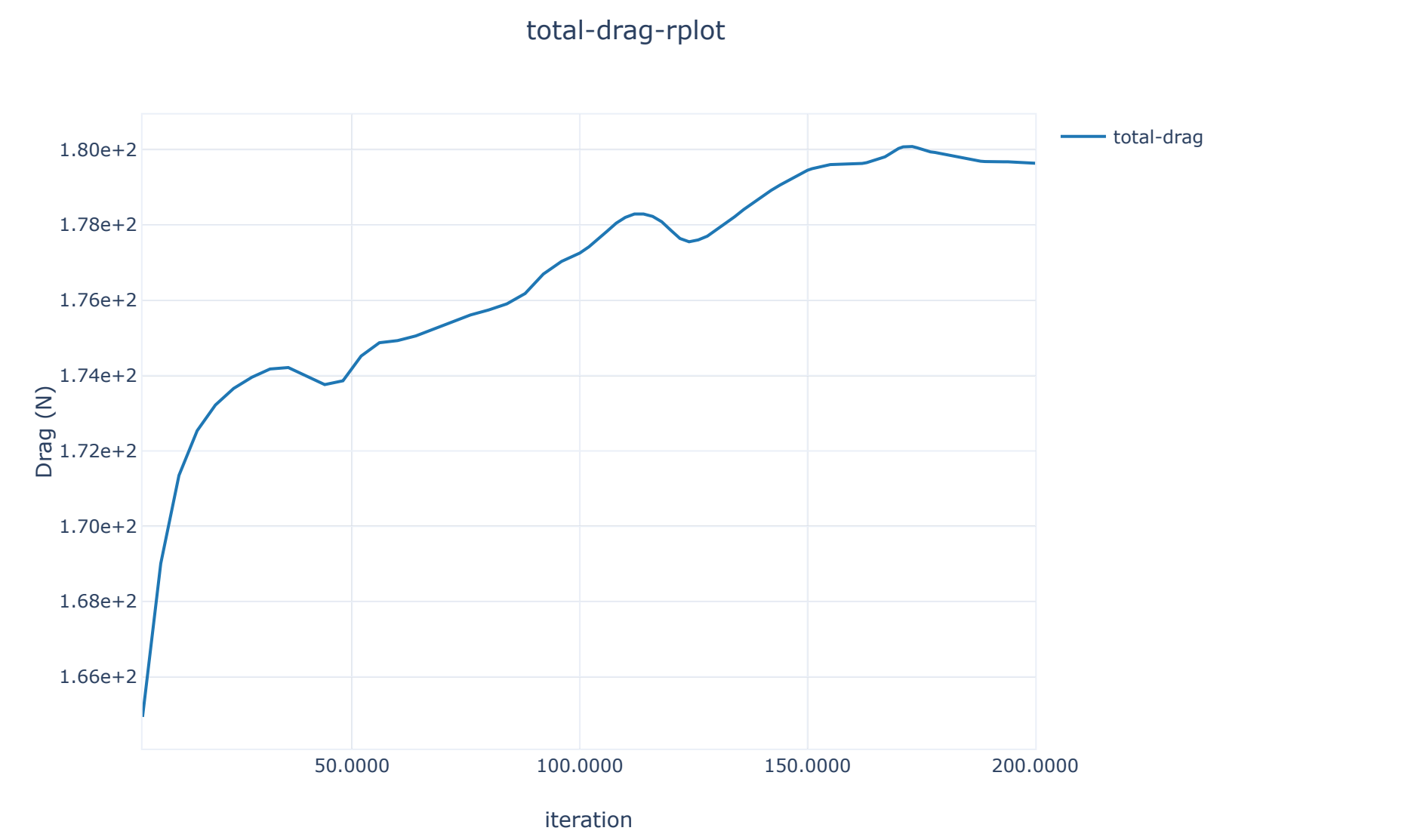
Residuals



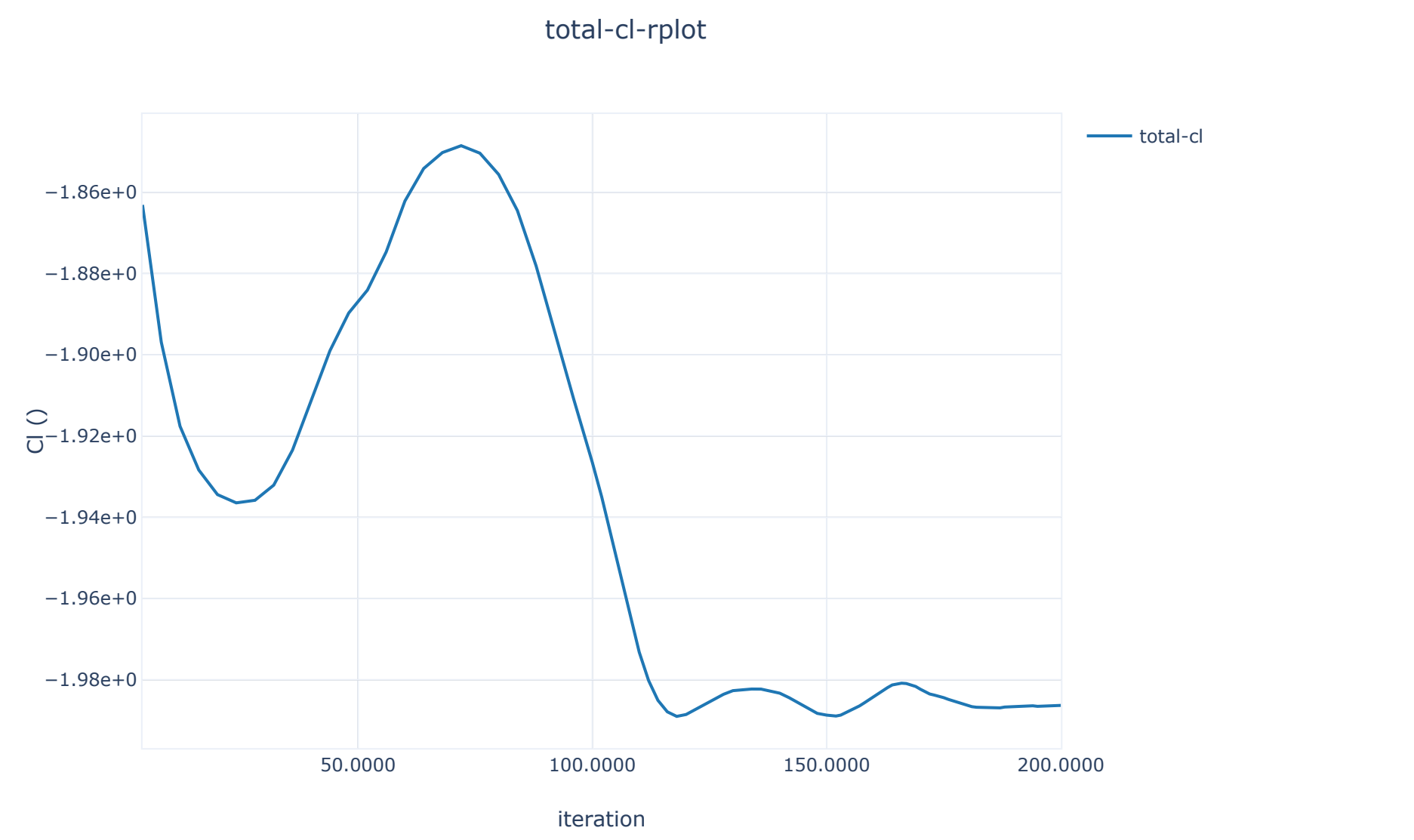
total-cd-rplot



total-drag-rplot

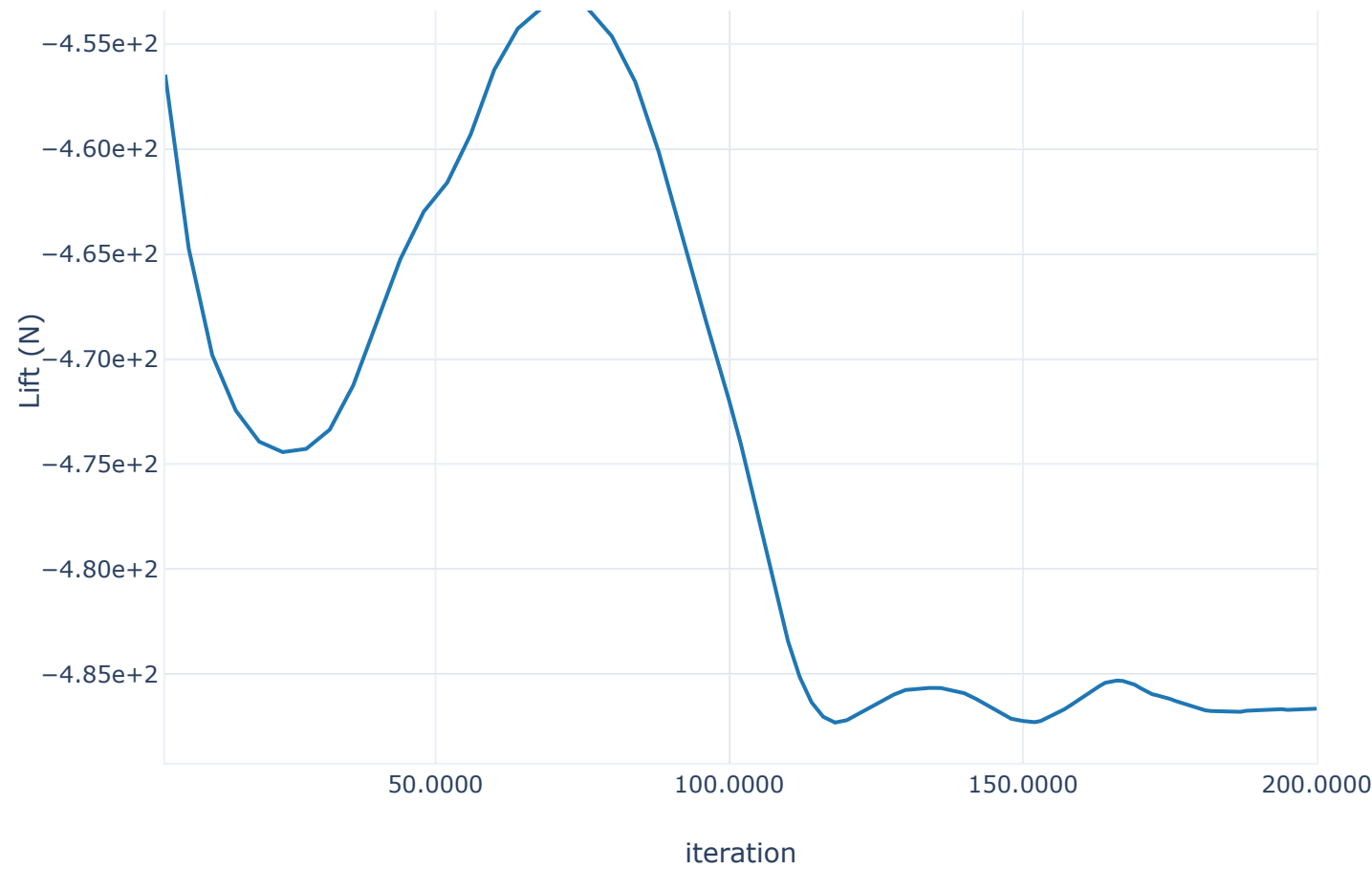


total-cl-rplot

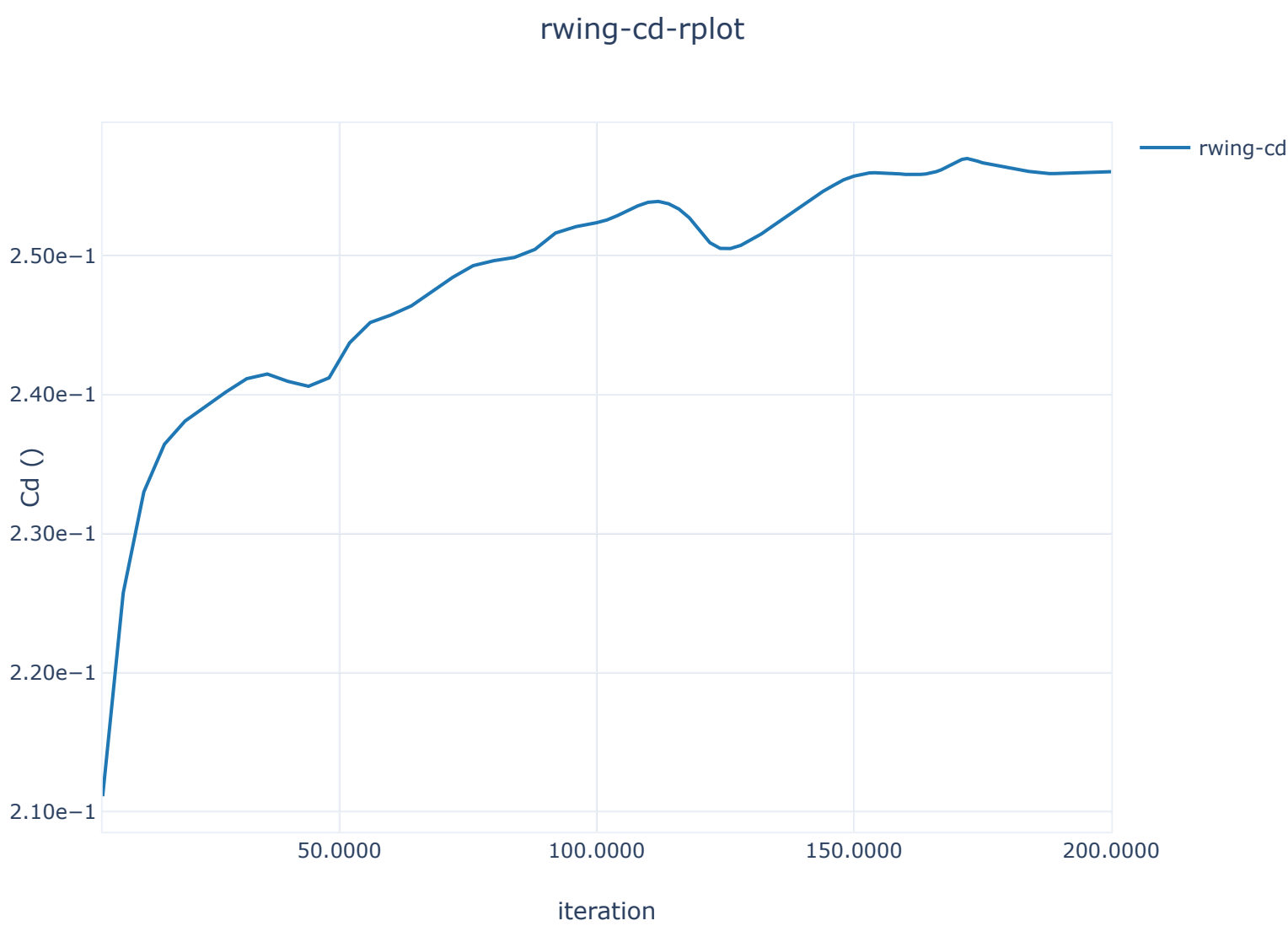


total-lift-rplot

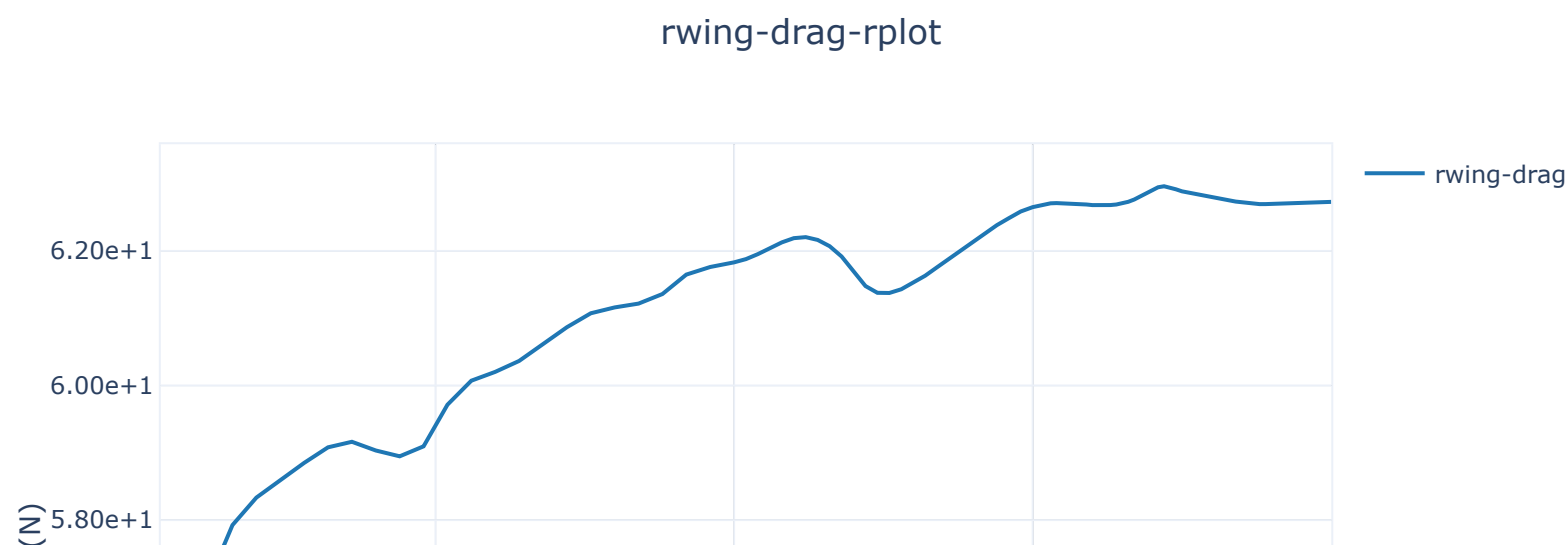


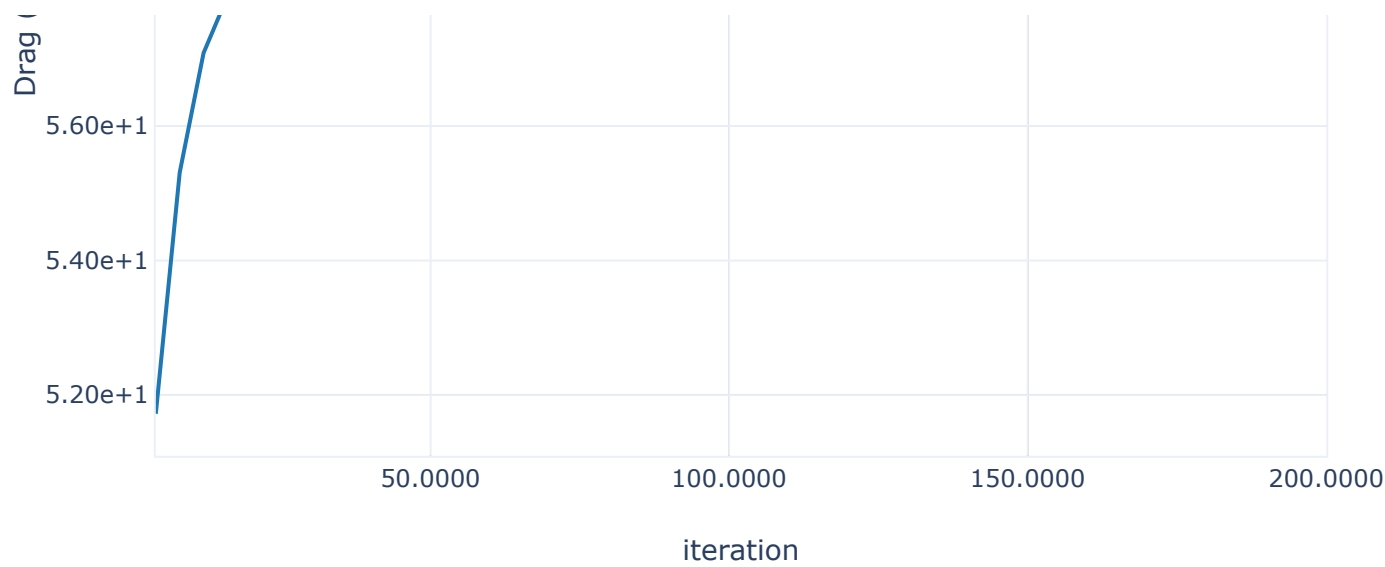


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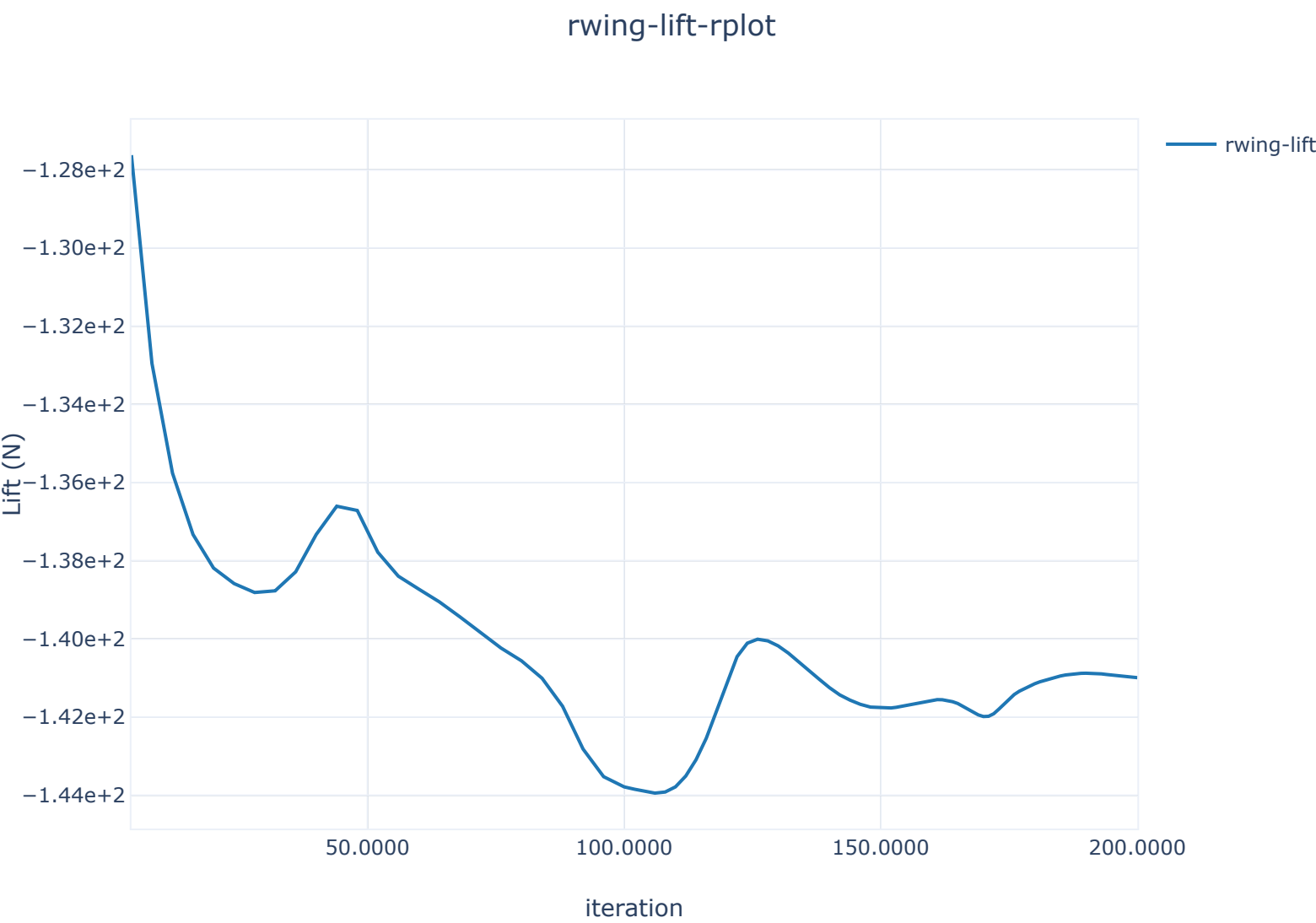


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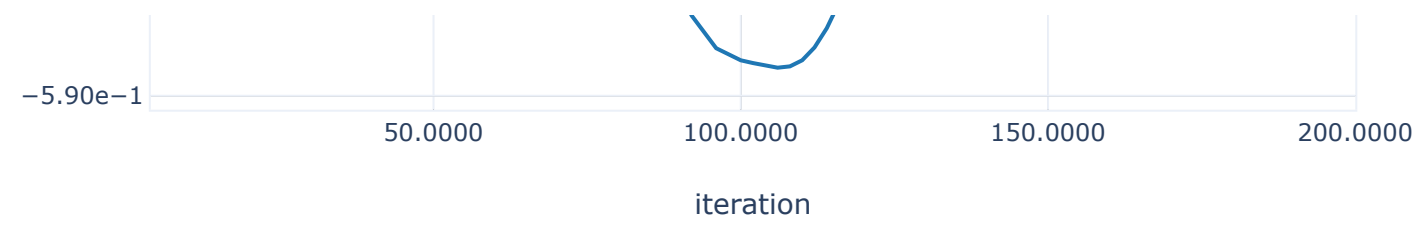


rwing-lift-rplot

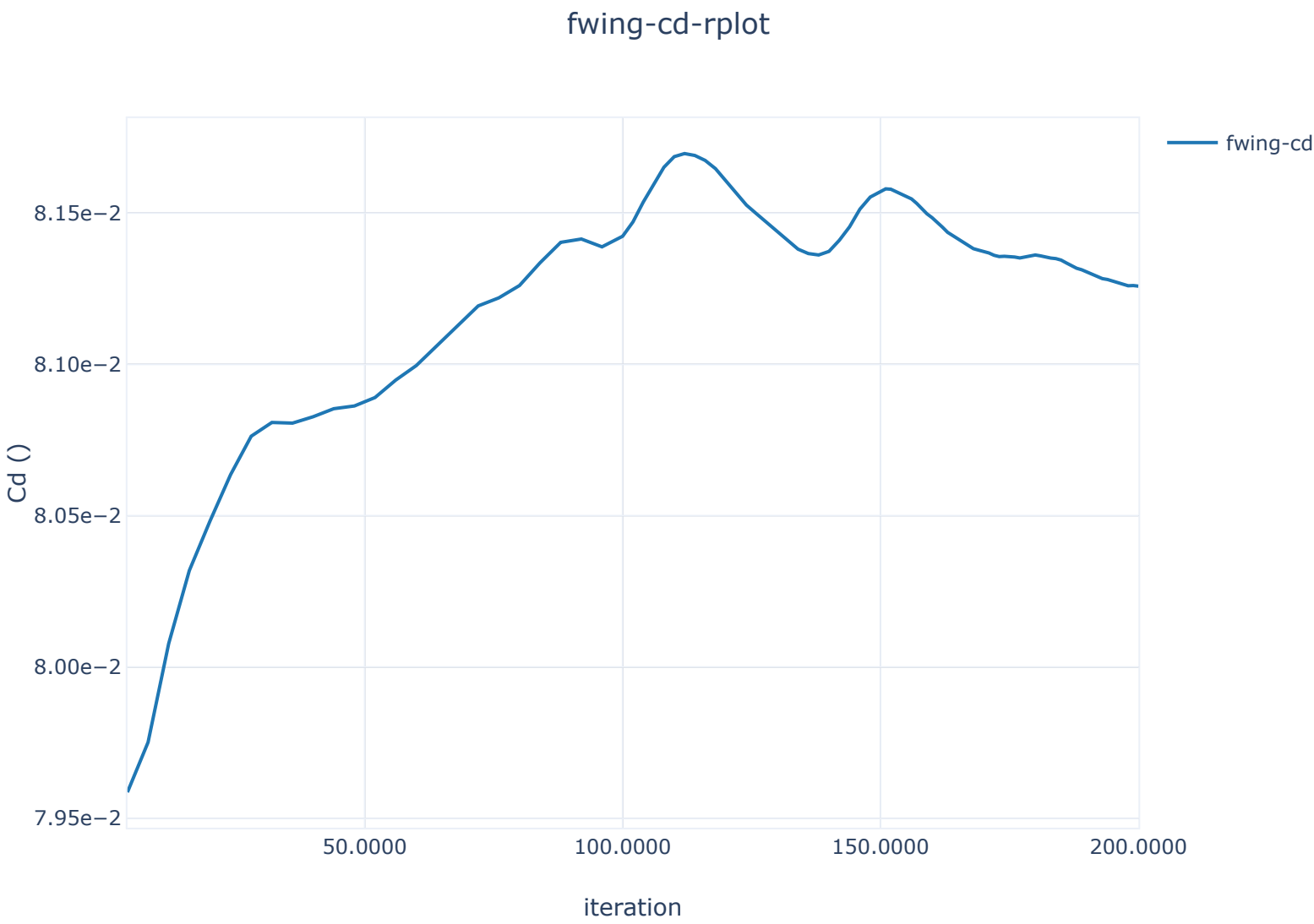


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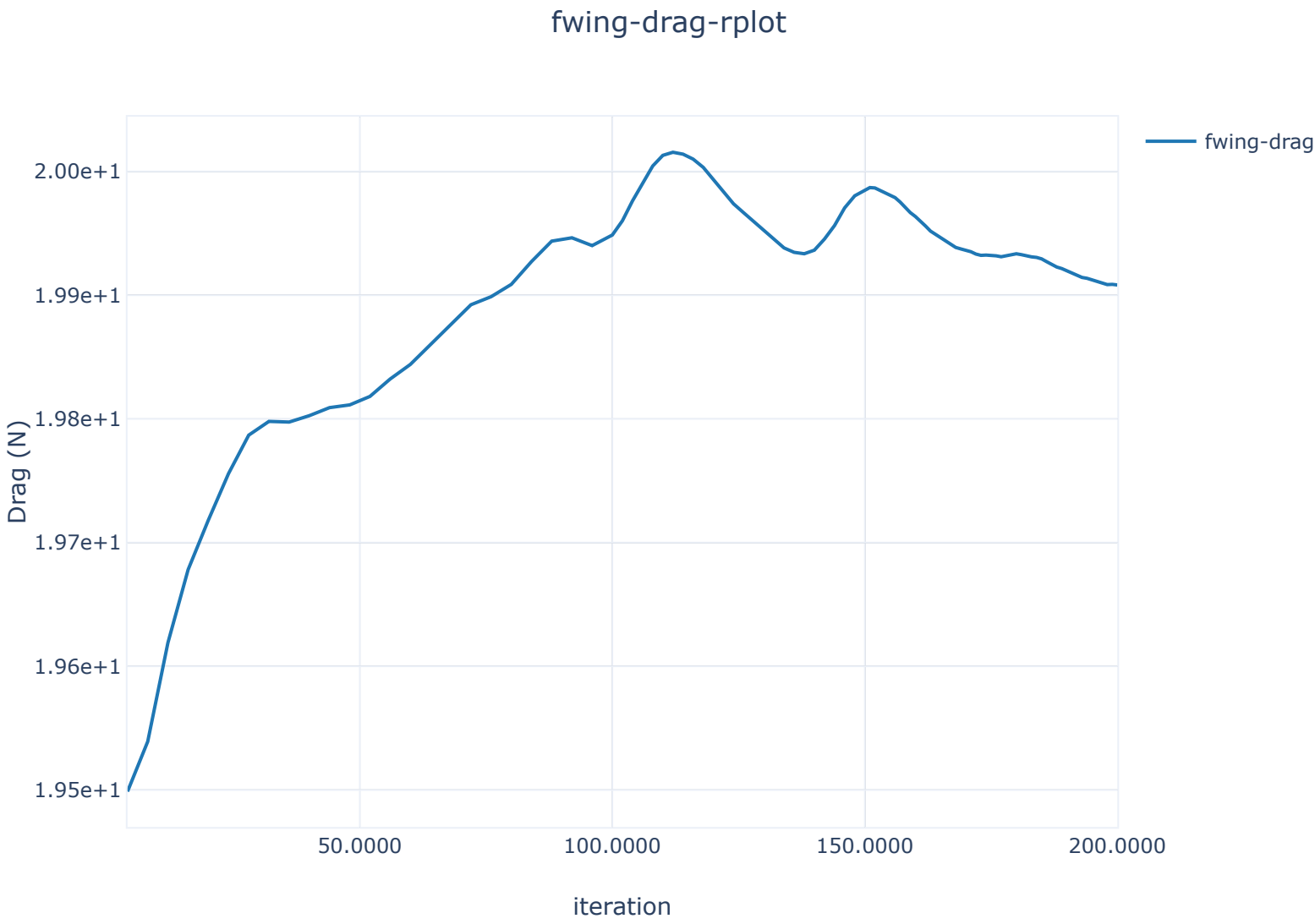




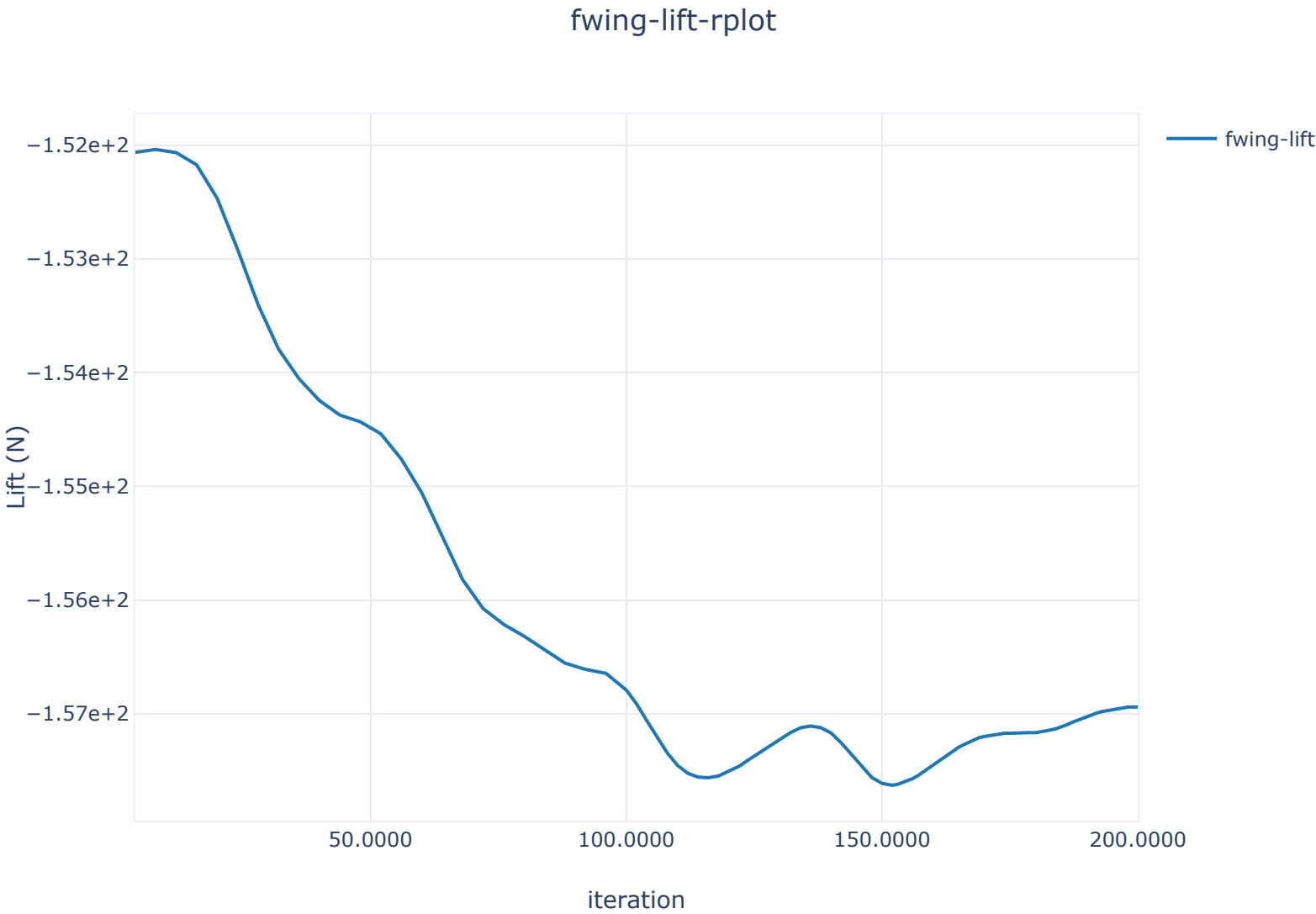
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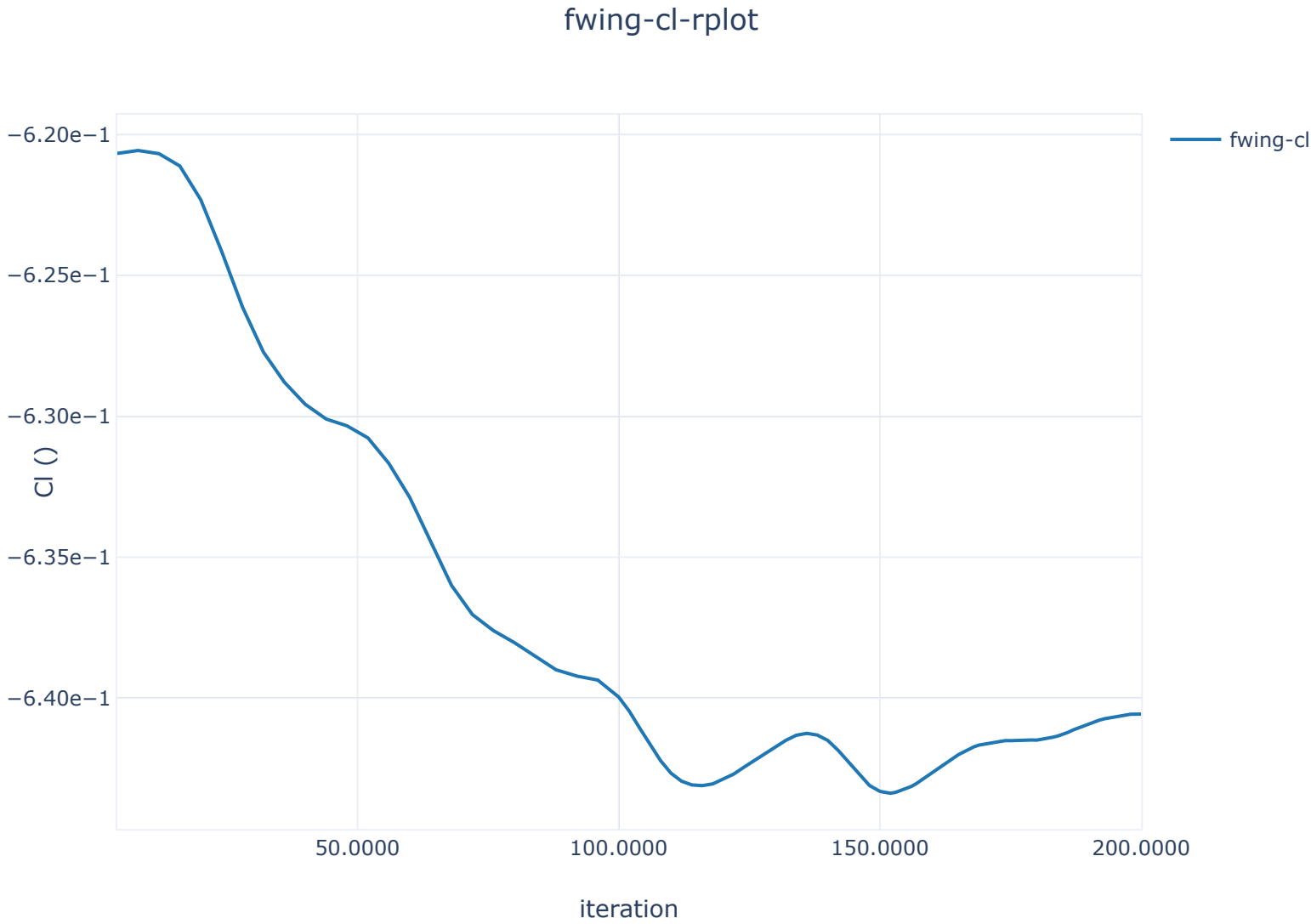
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fwing-lift-rplot

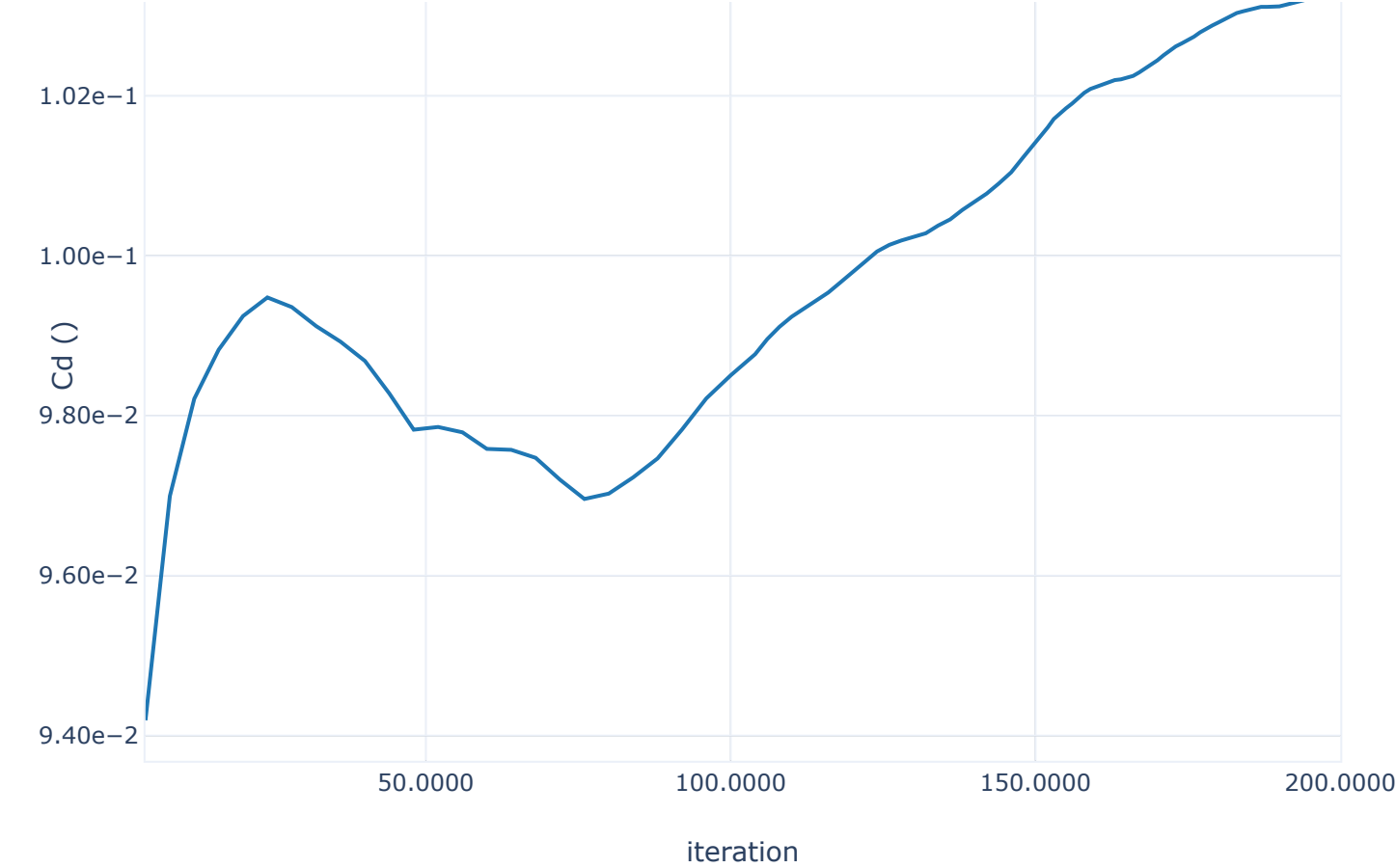


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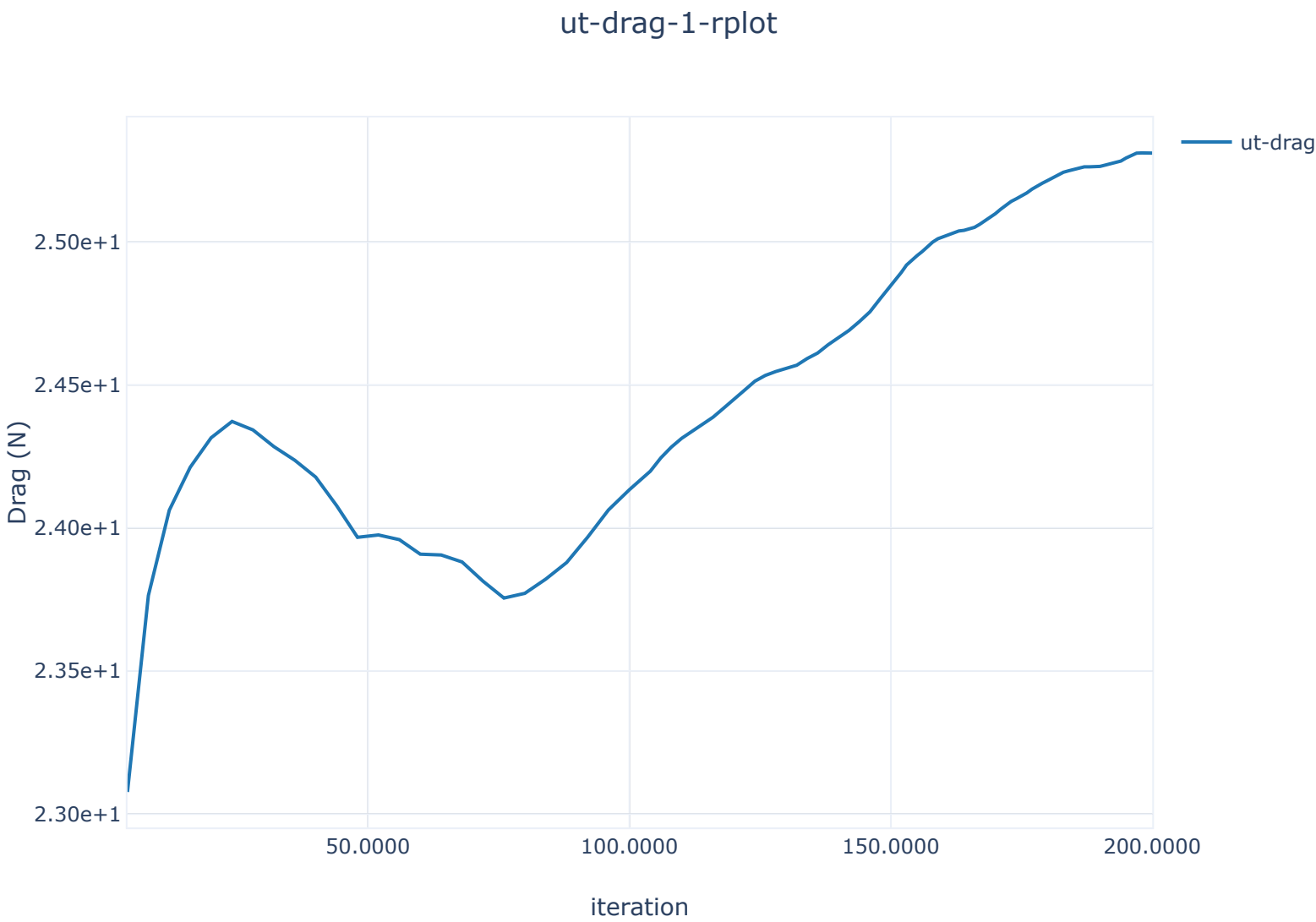


ut-cd-rplot

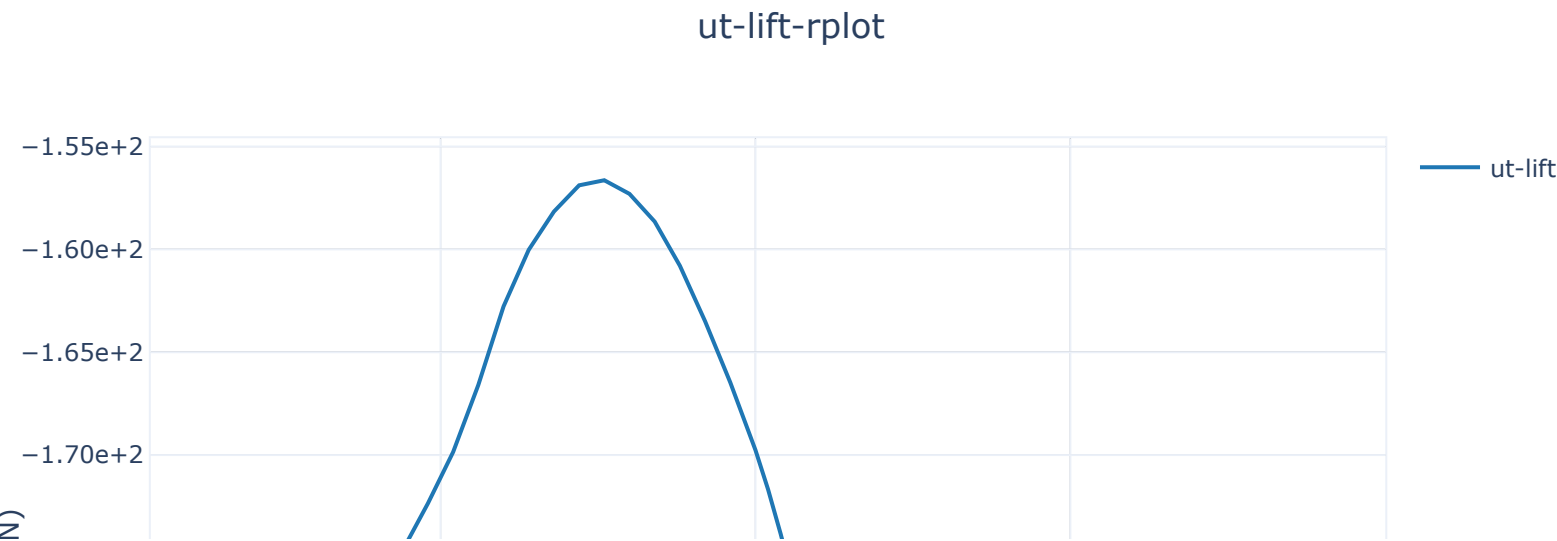


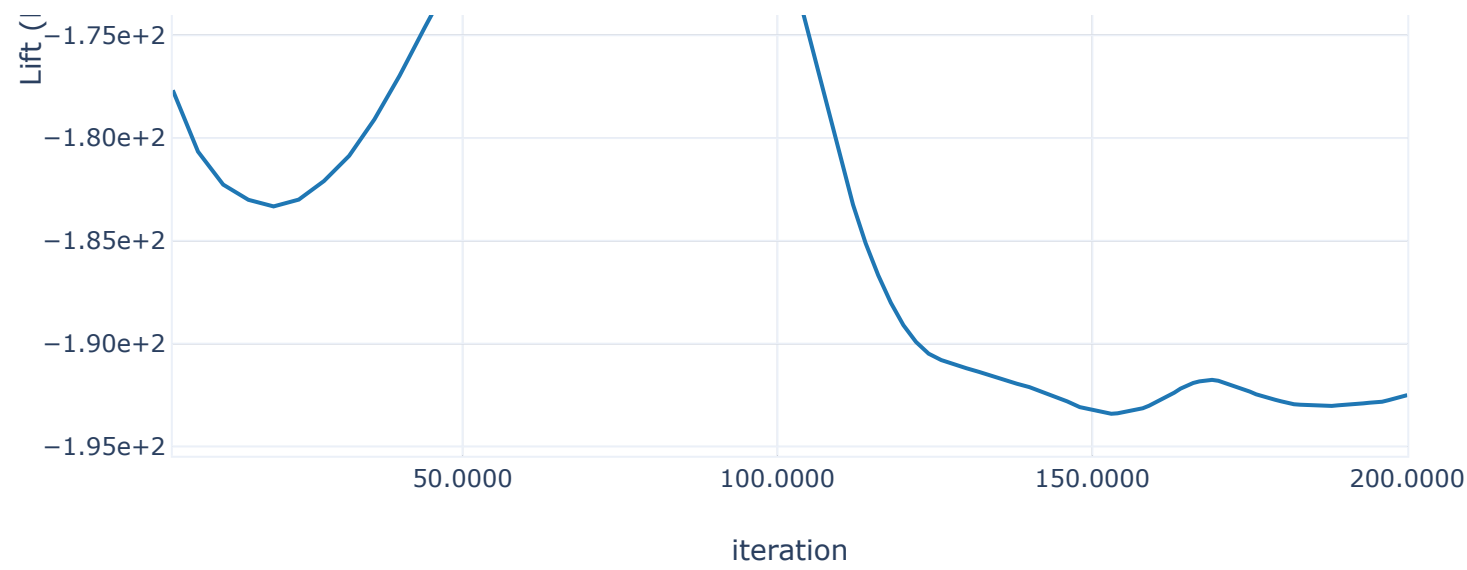


ut-drag-1-rplot

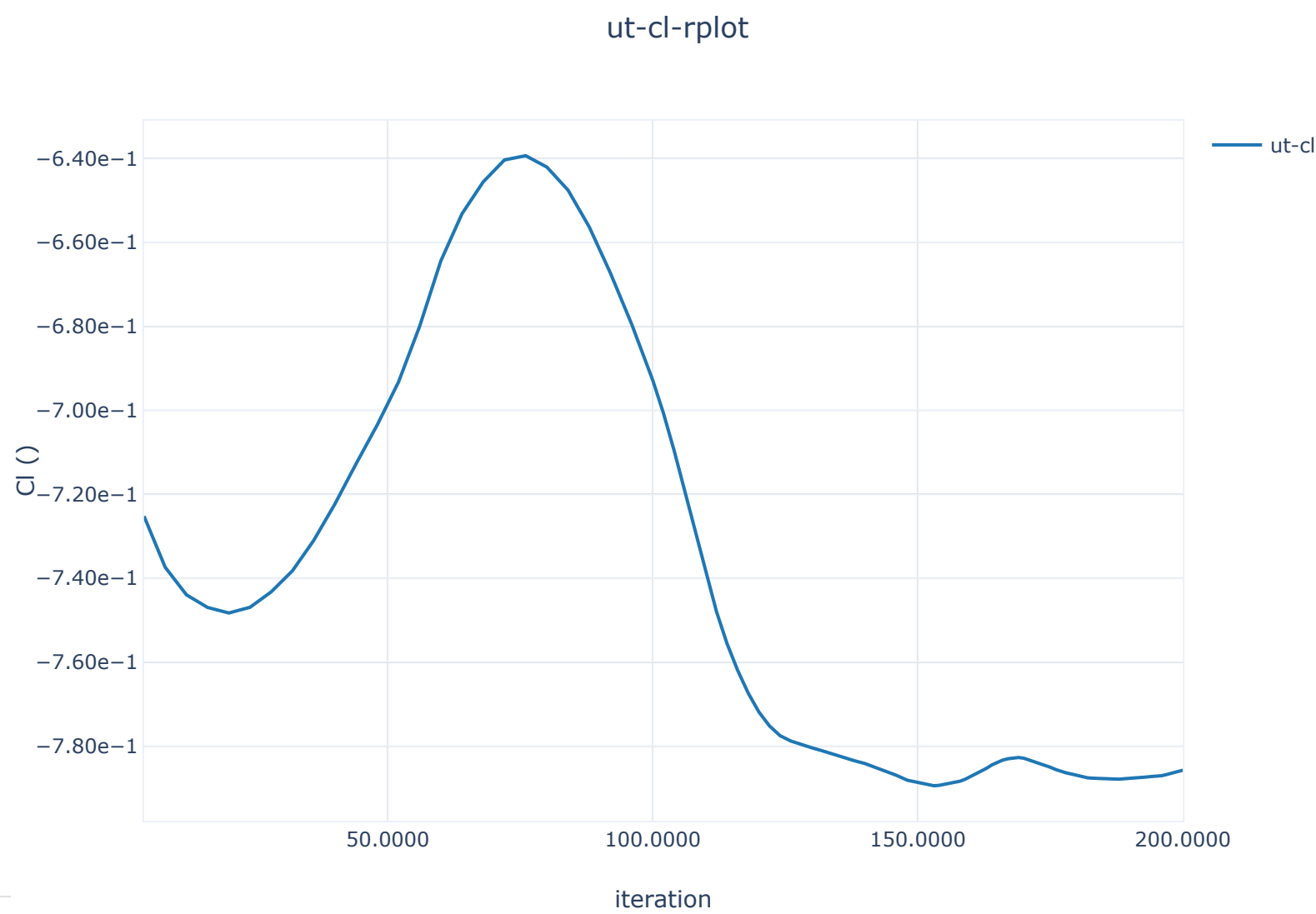


ut-lift-rplot



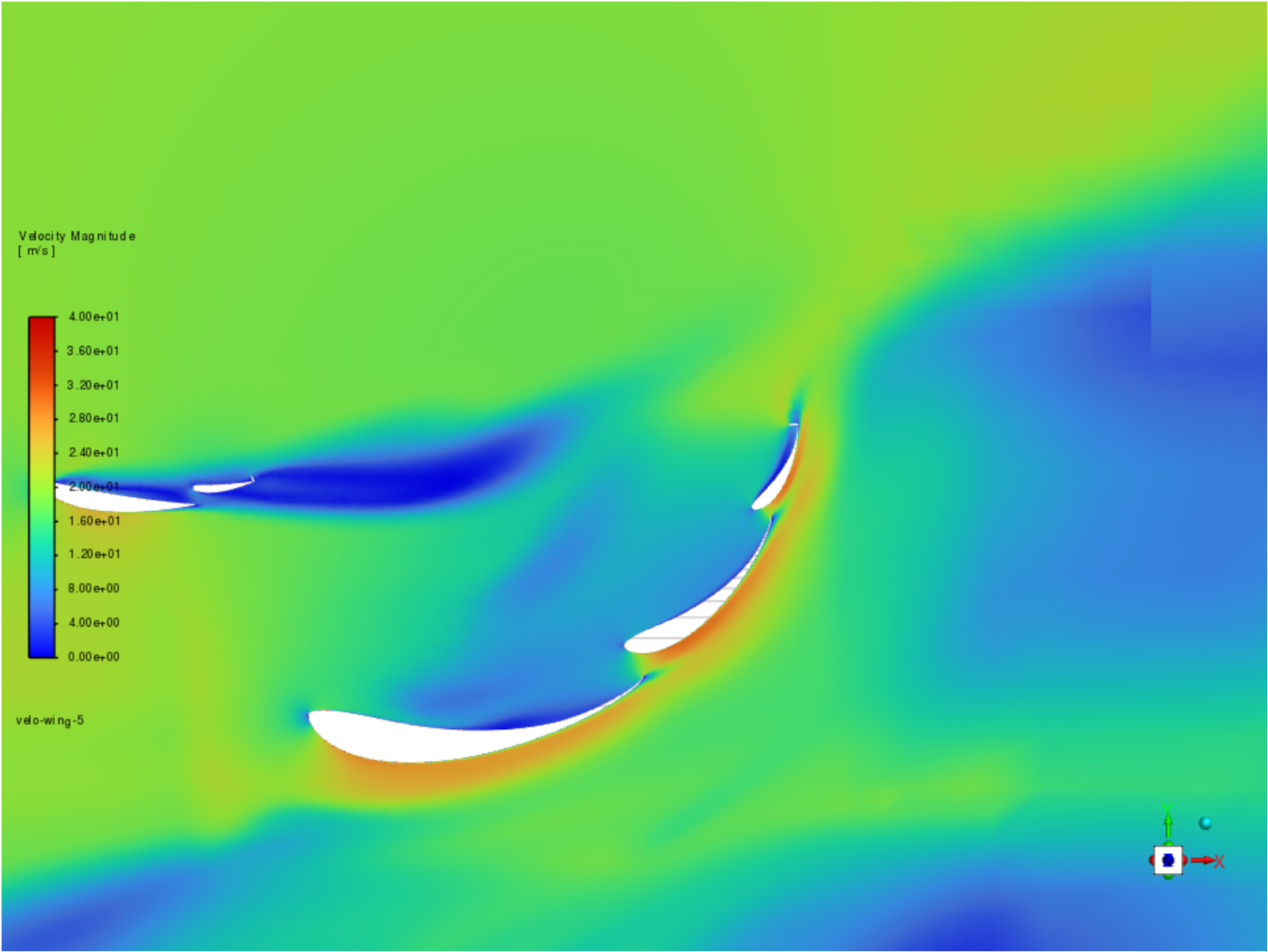


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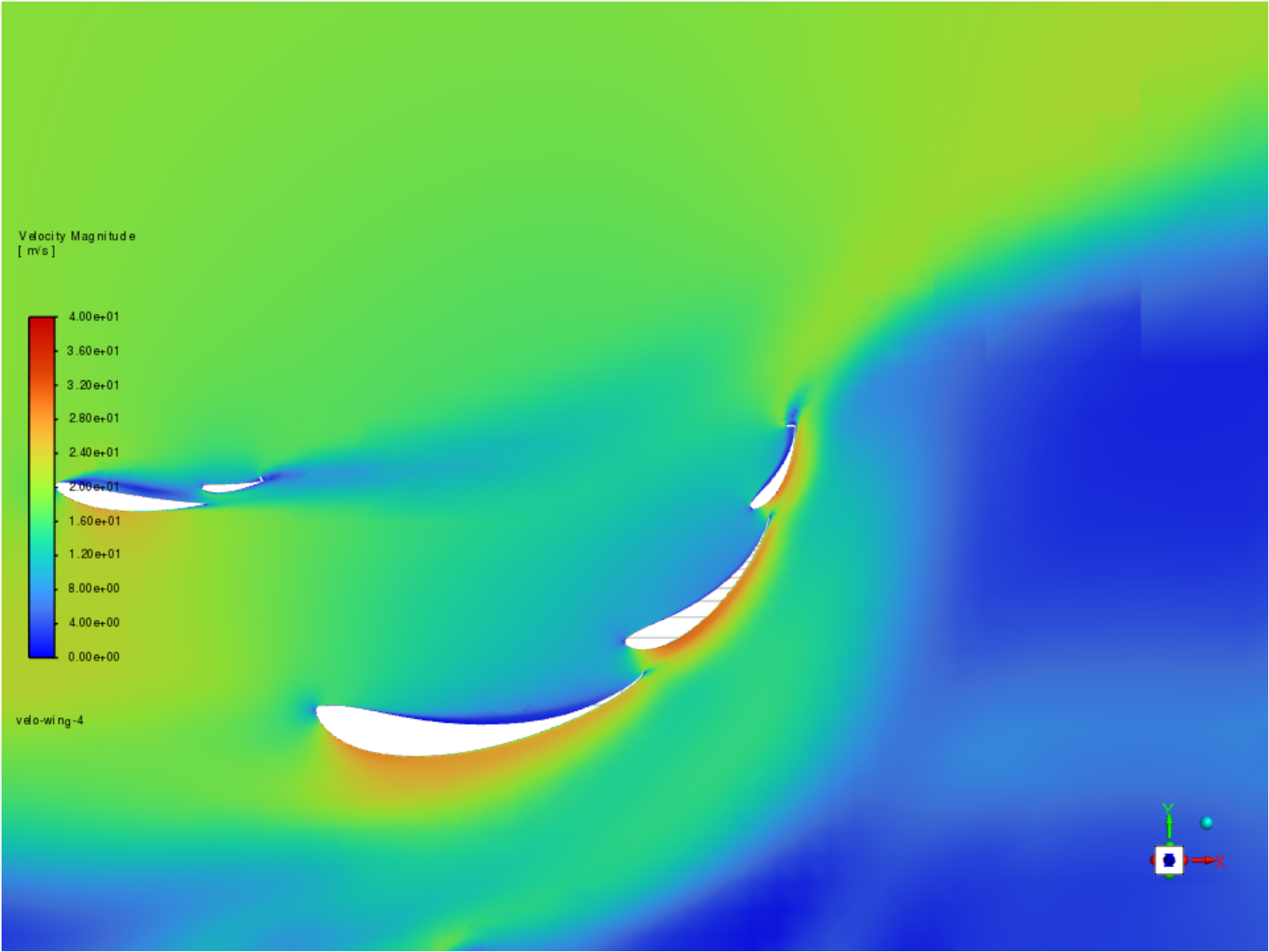


Contours

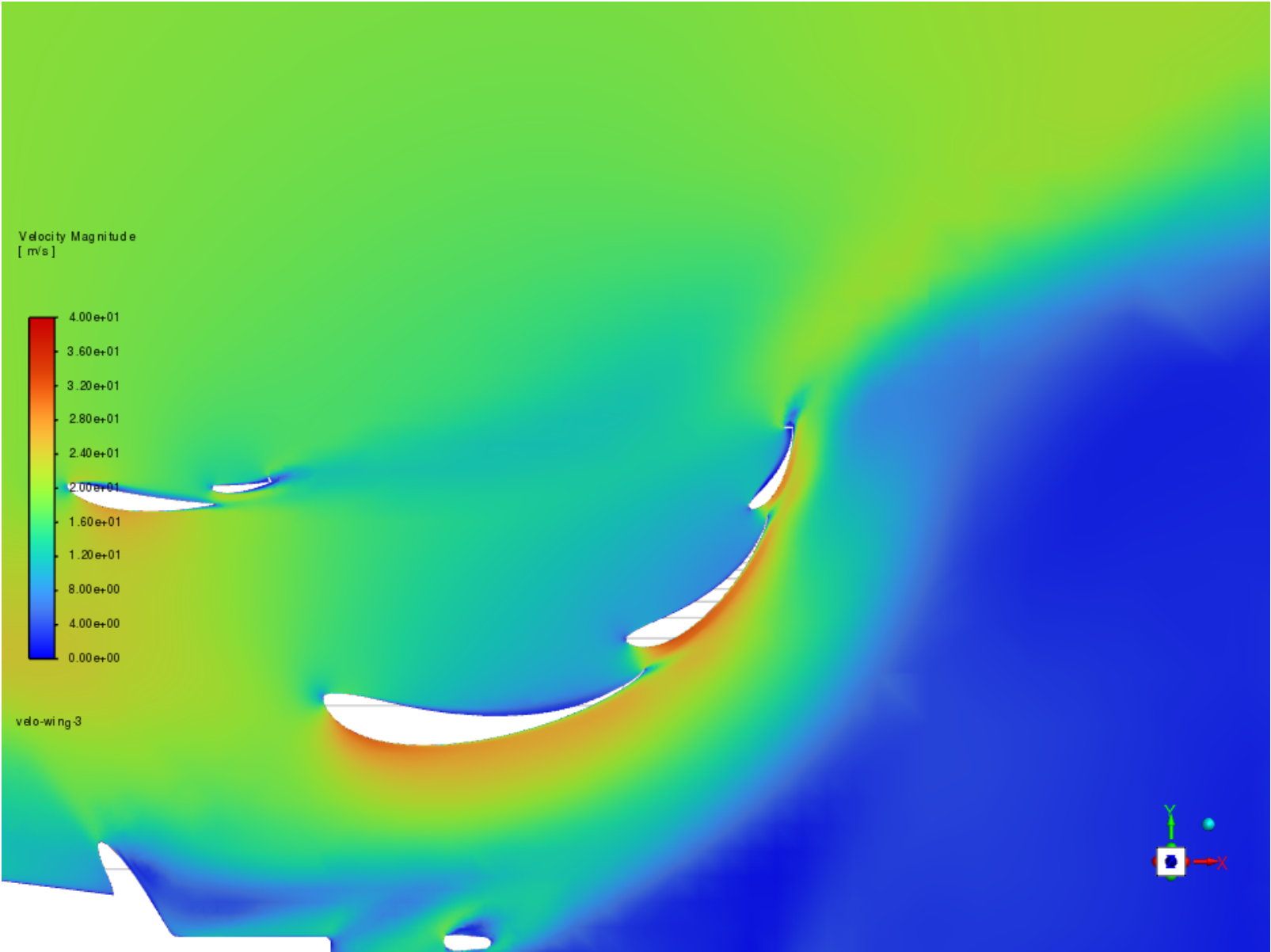
velo-wing-5



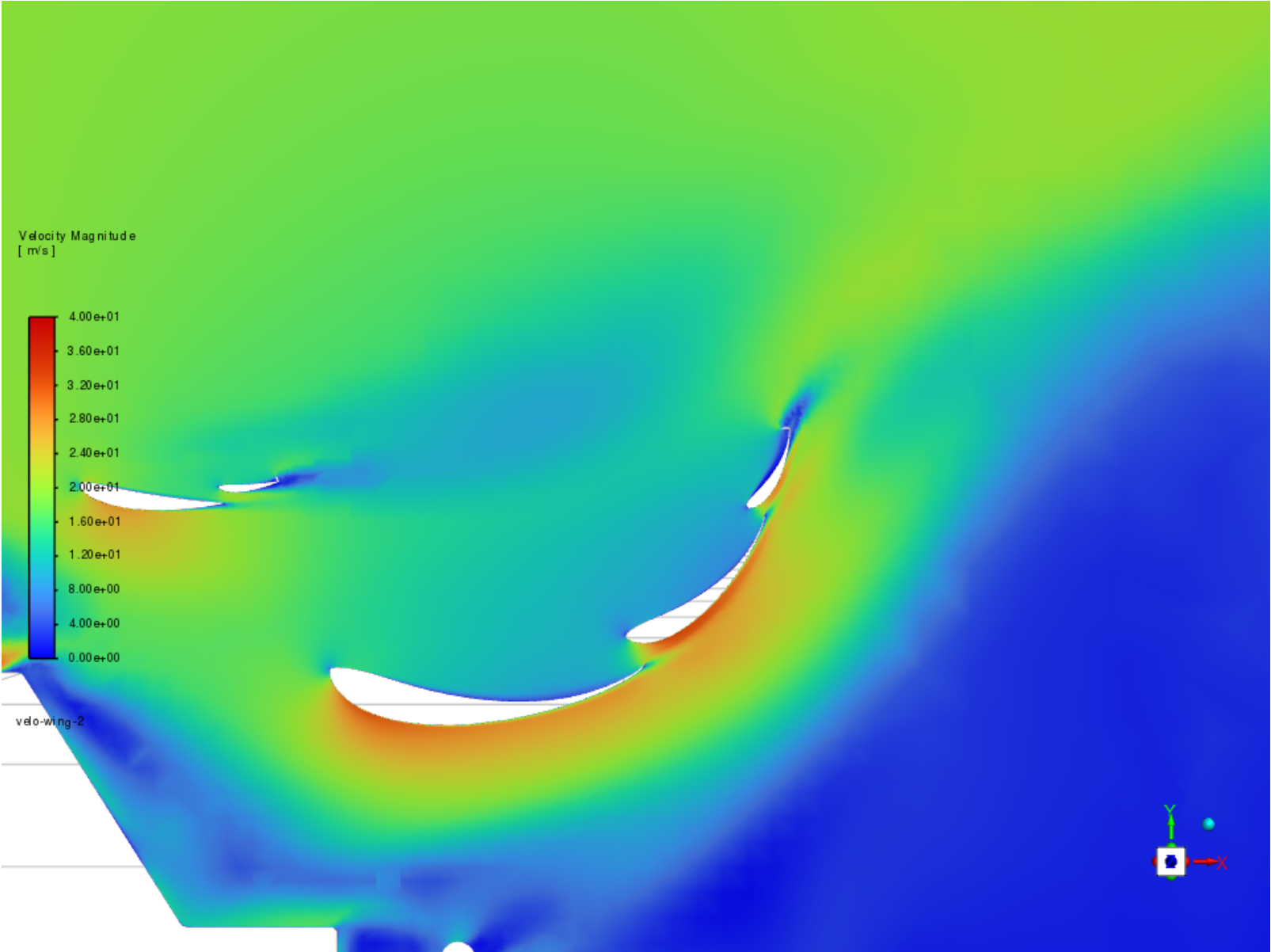
velo-wing-4



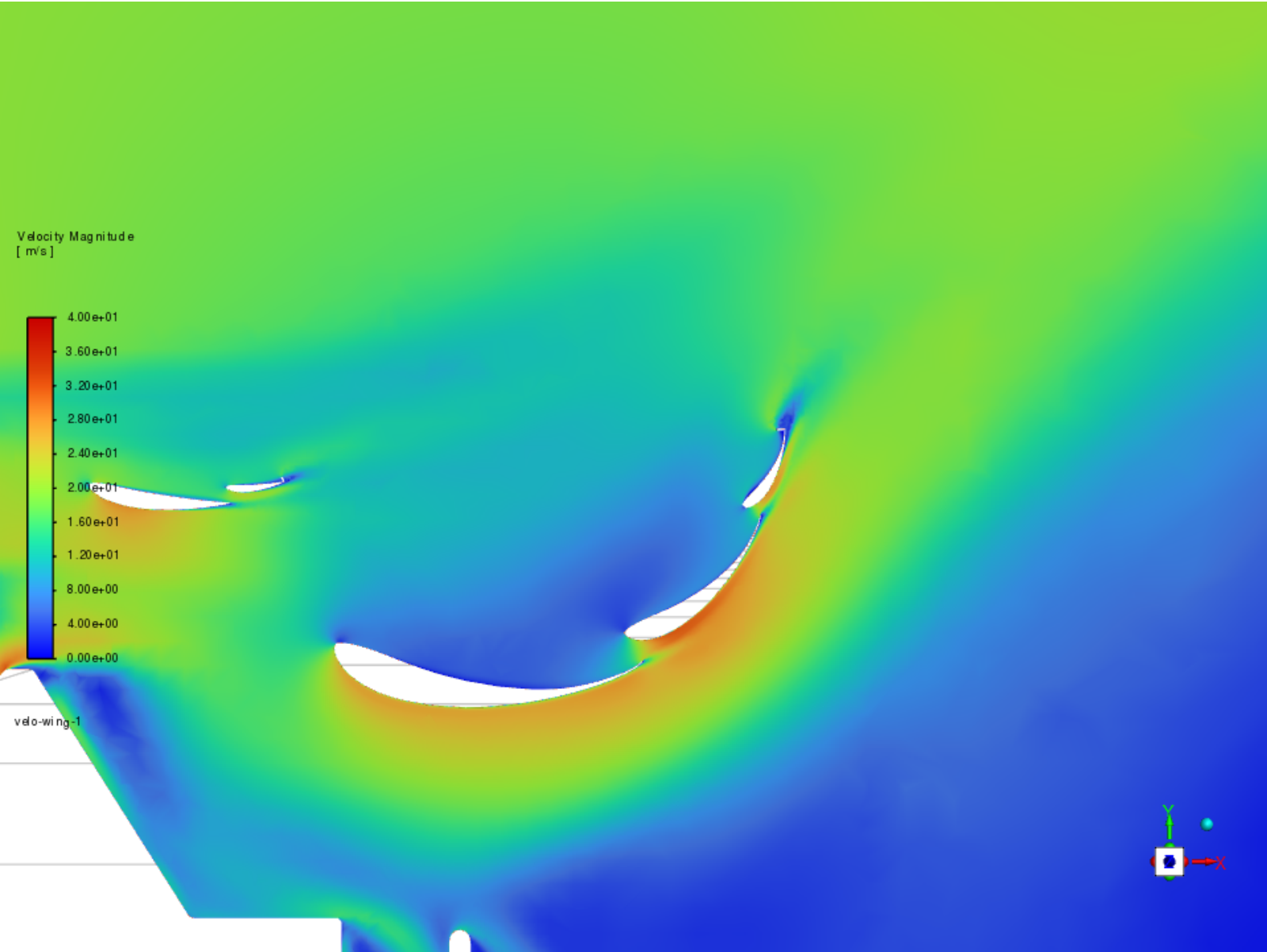
velo-wing-3



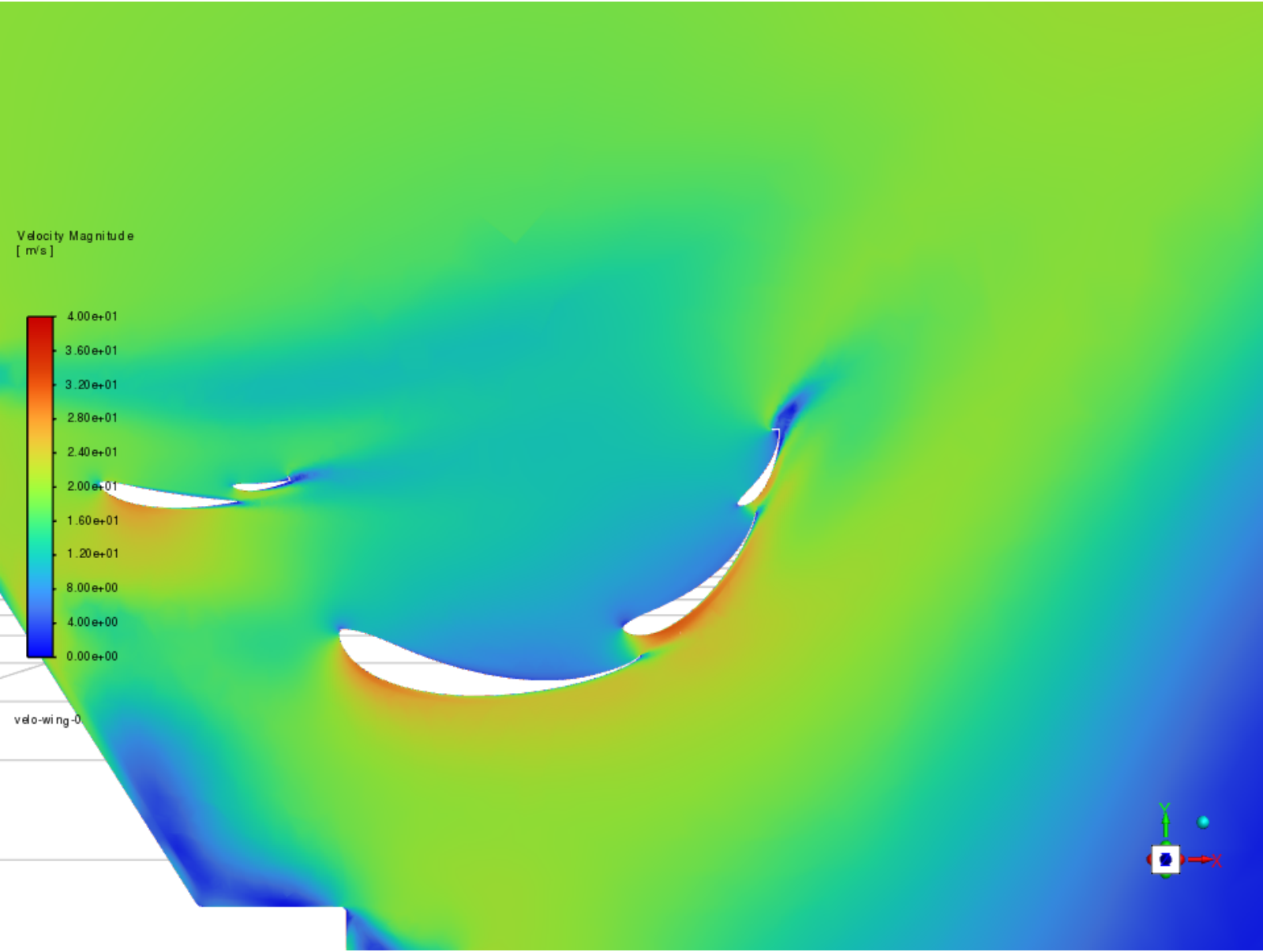
velo-wing-2



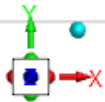
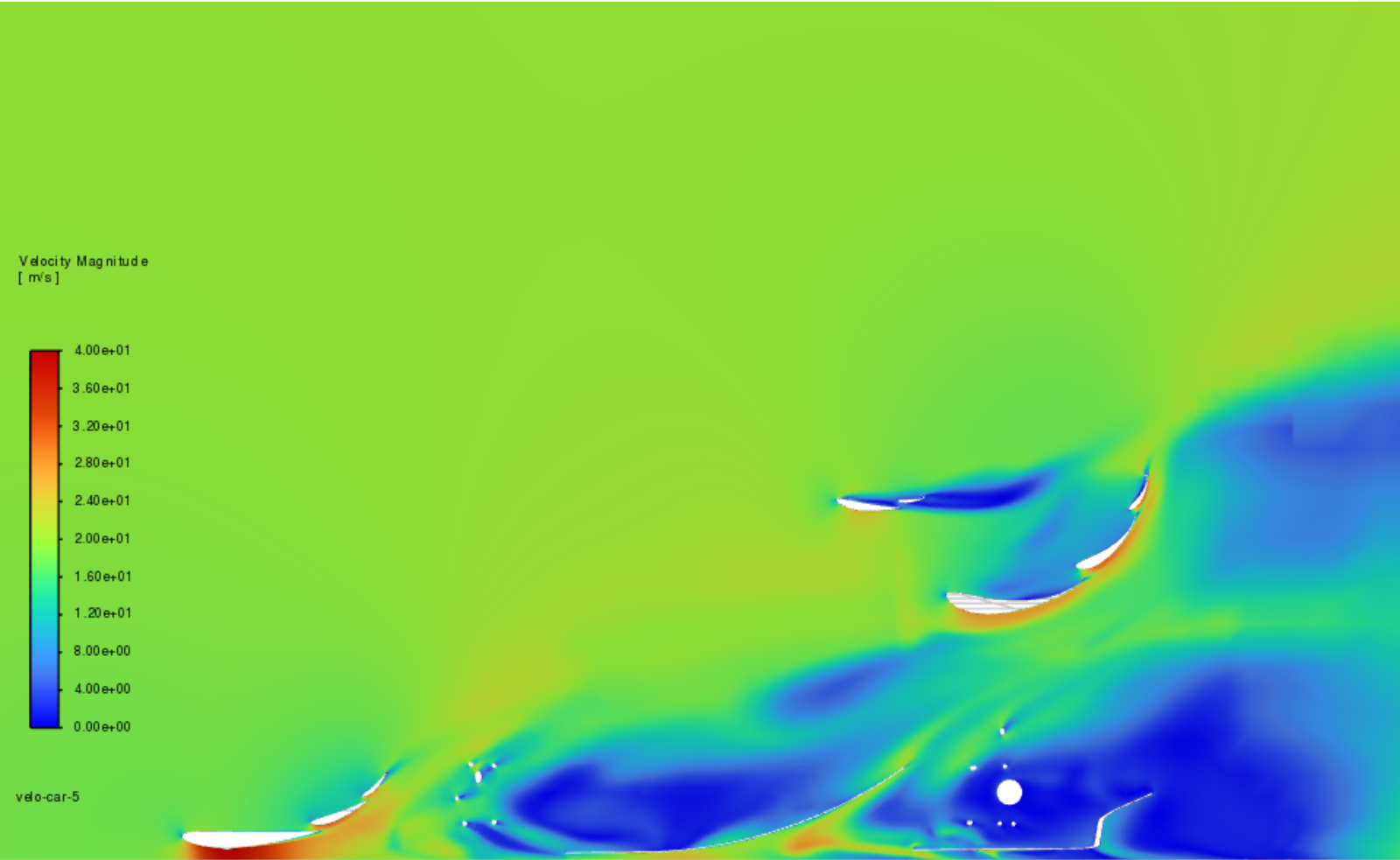
velo-wing-1



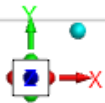
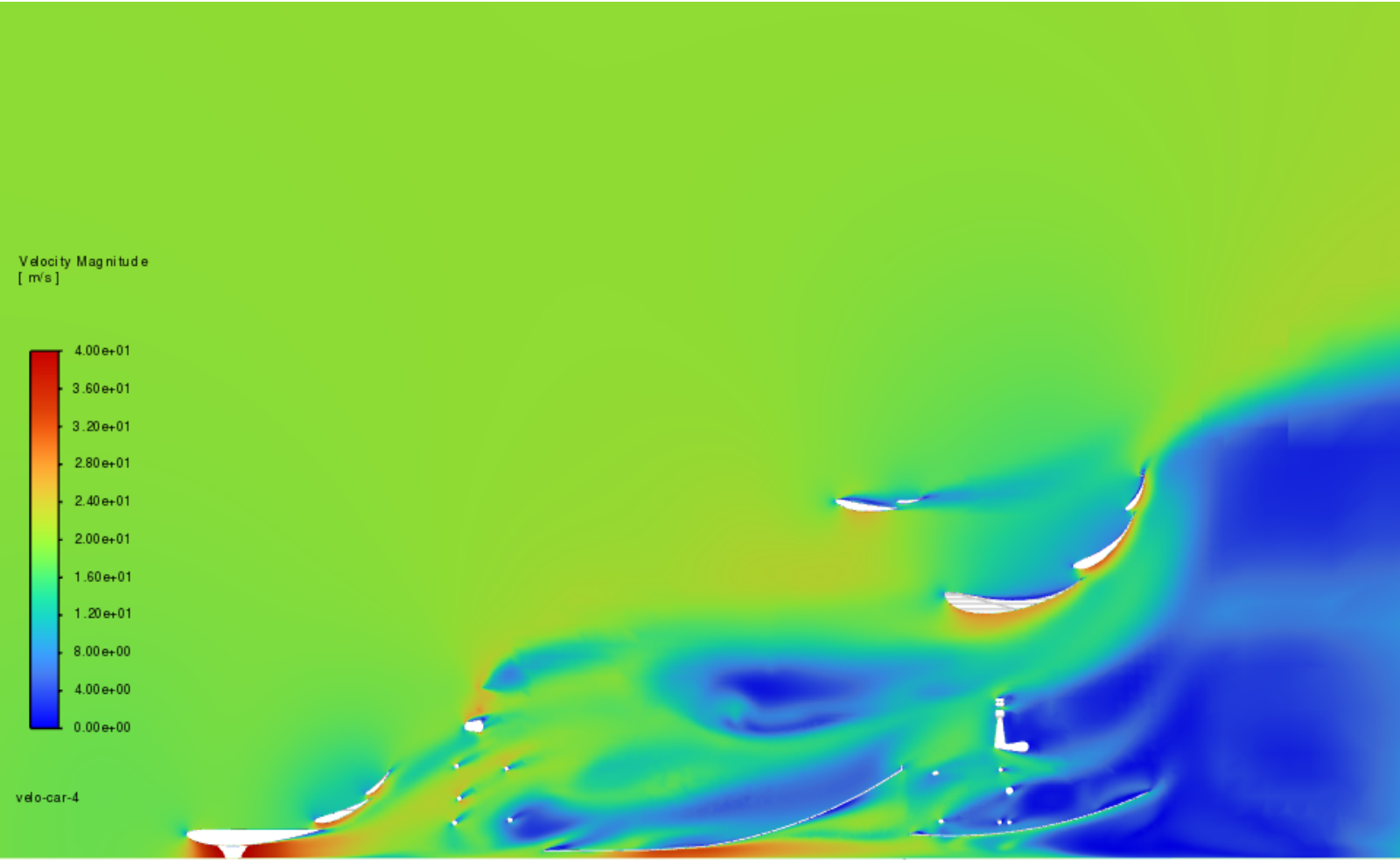
velo-wing-0



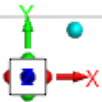
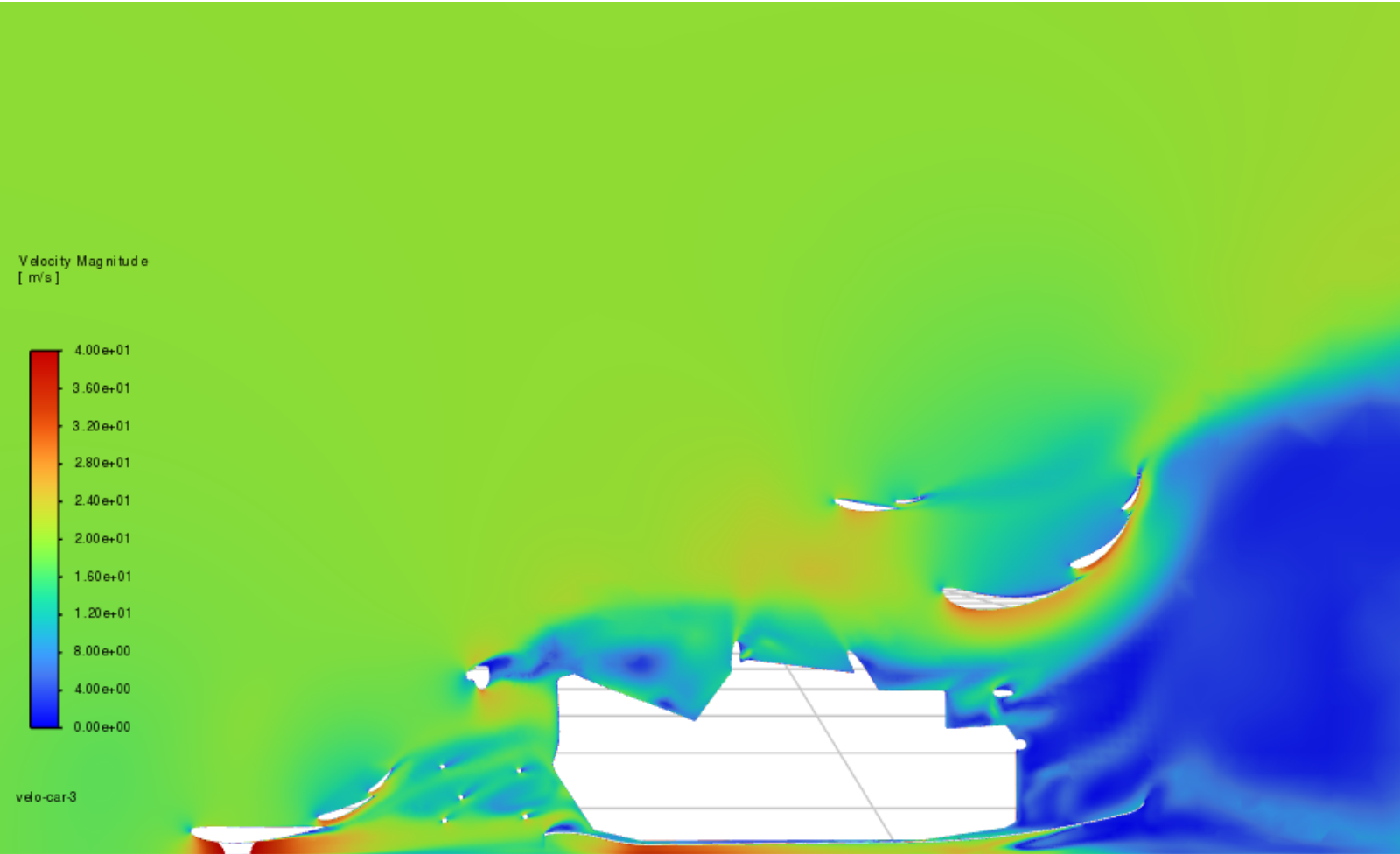
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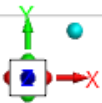
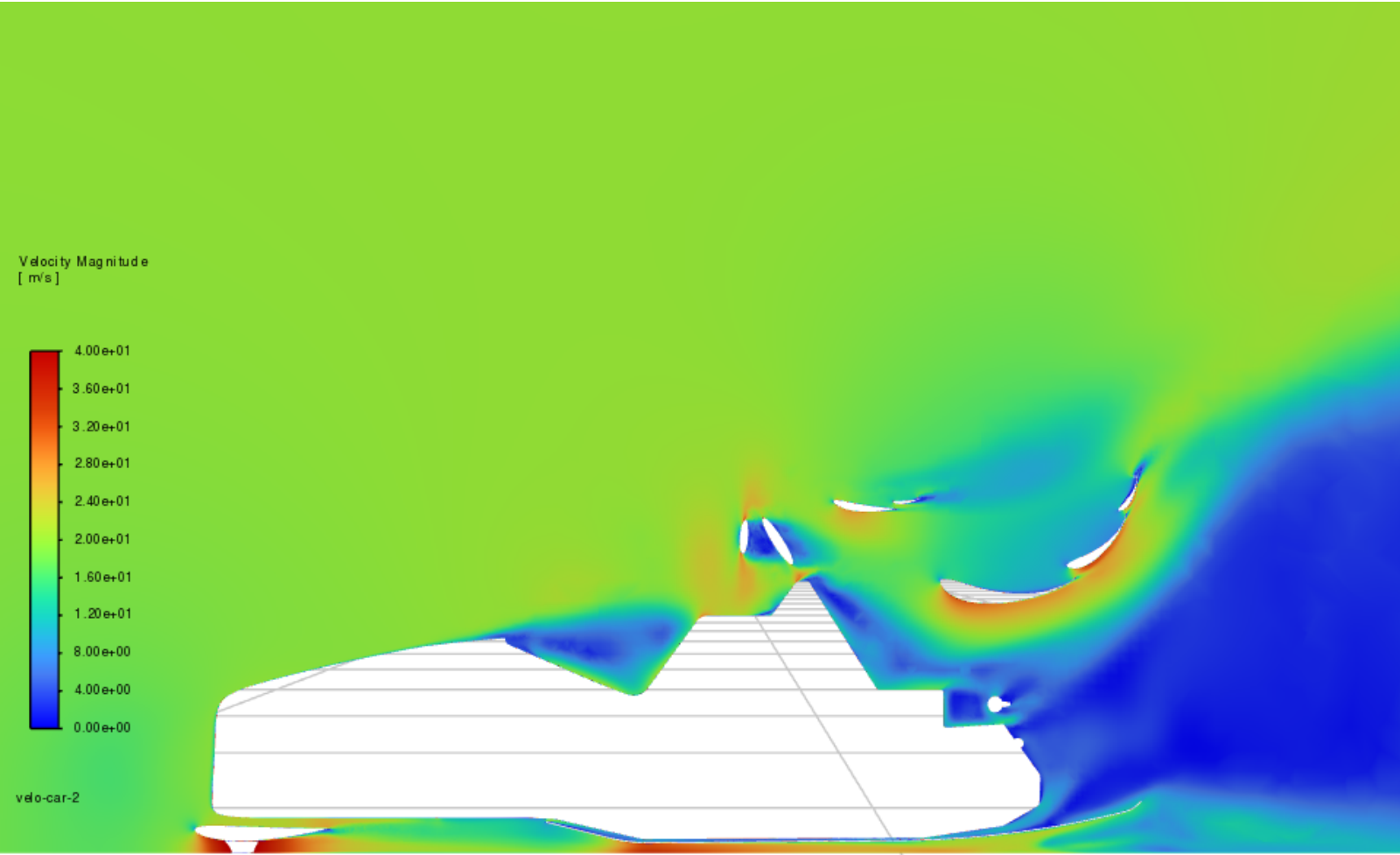
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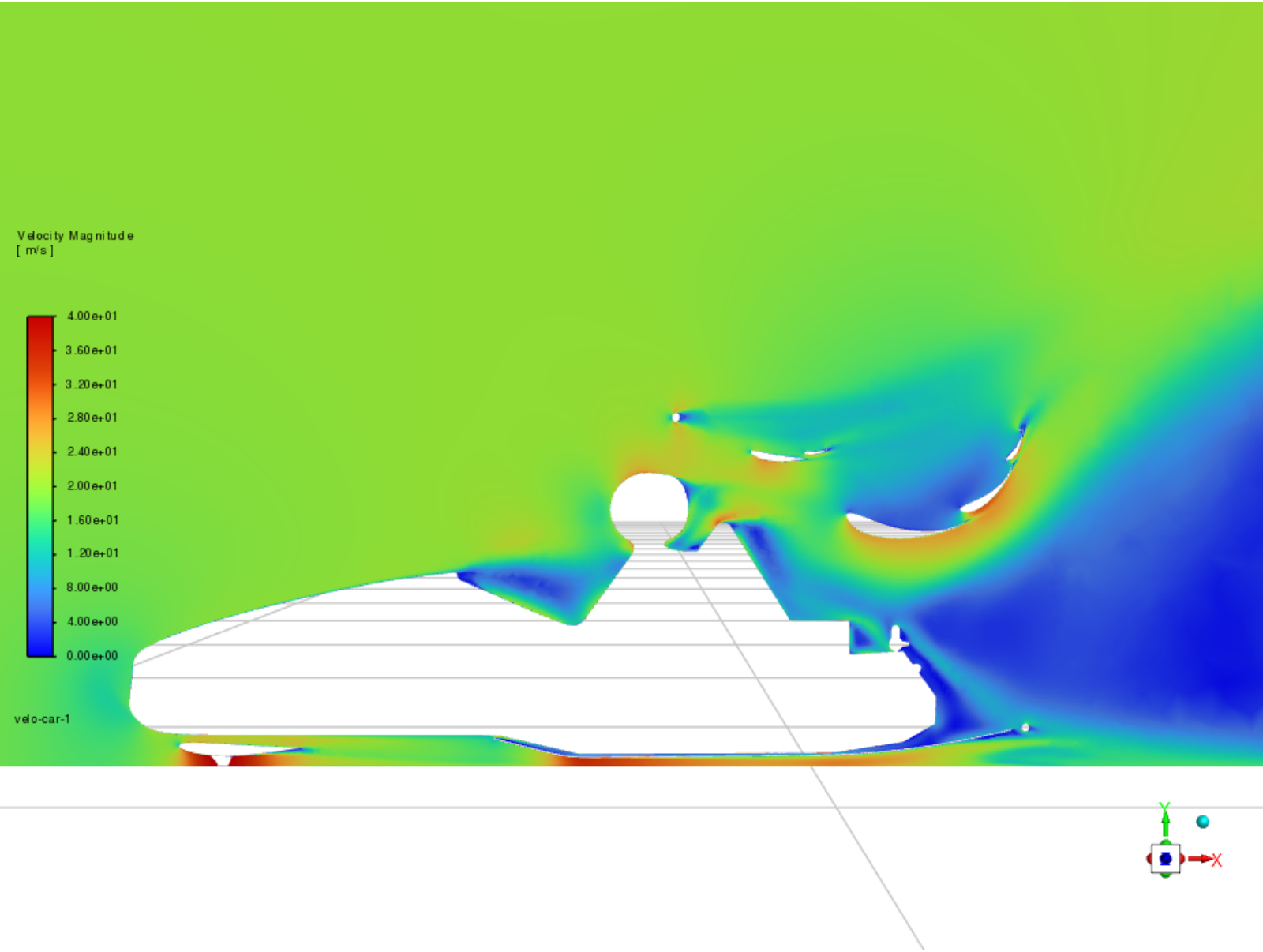
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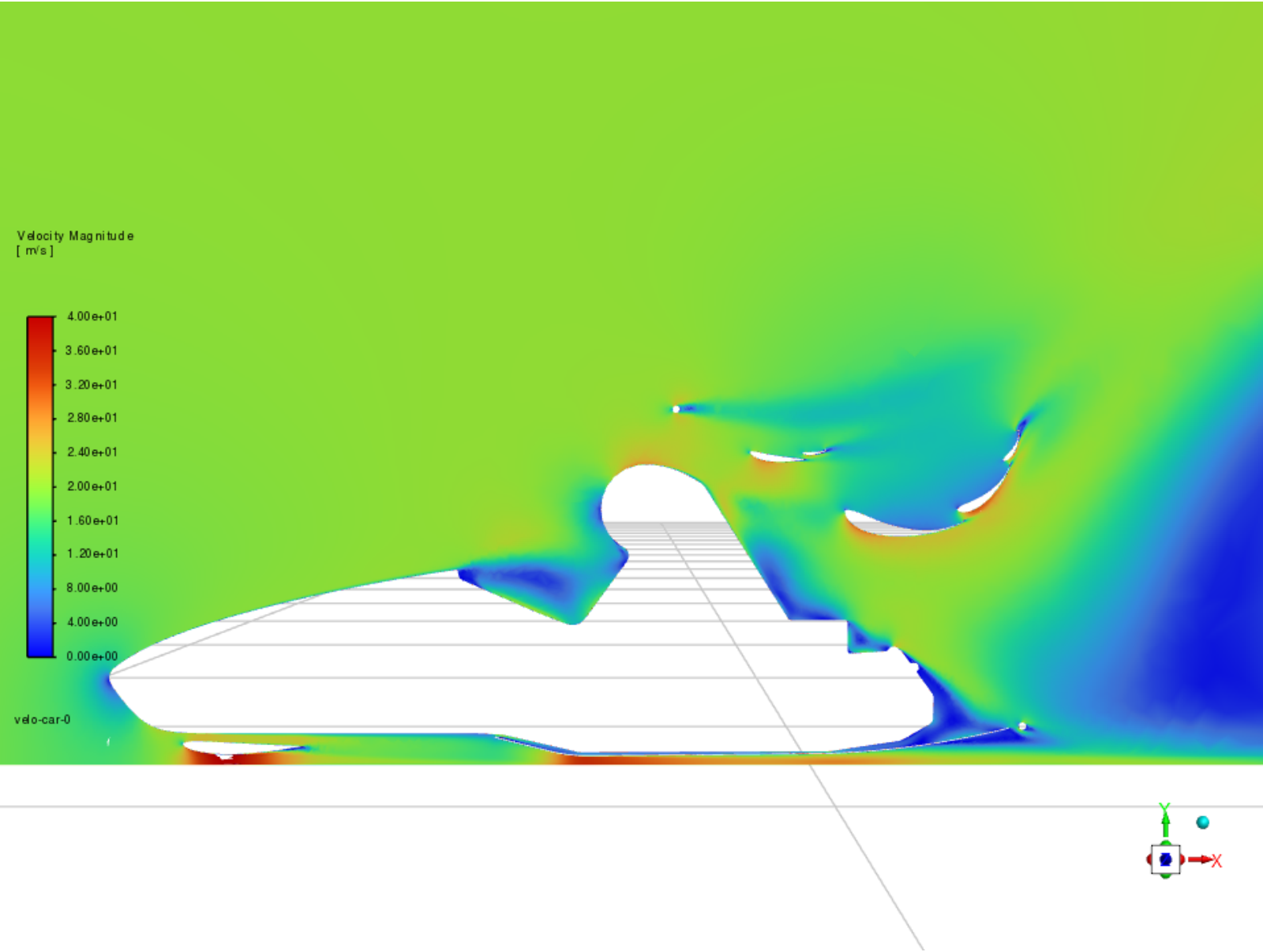
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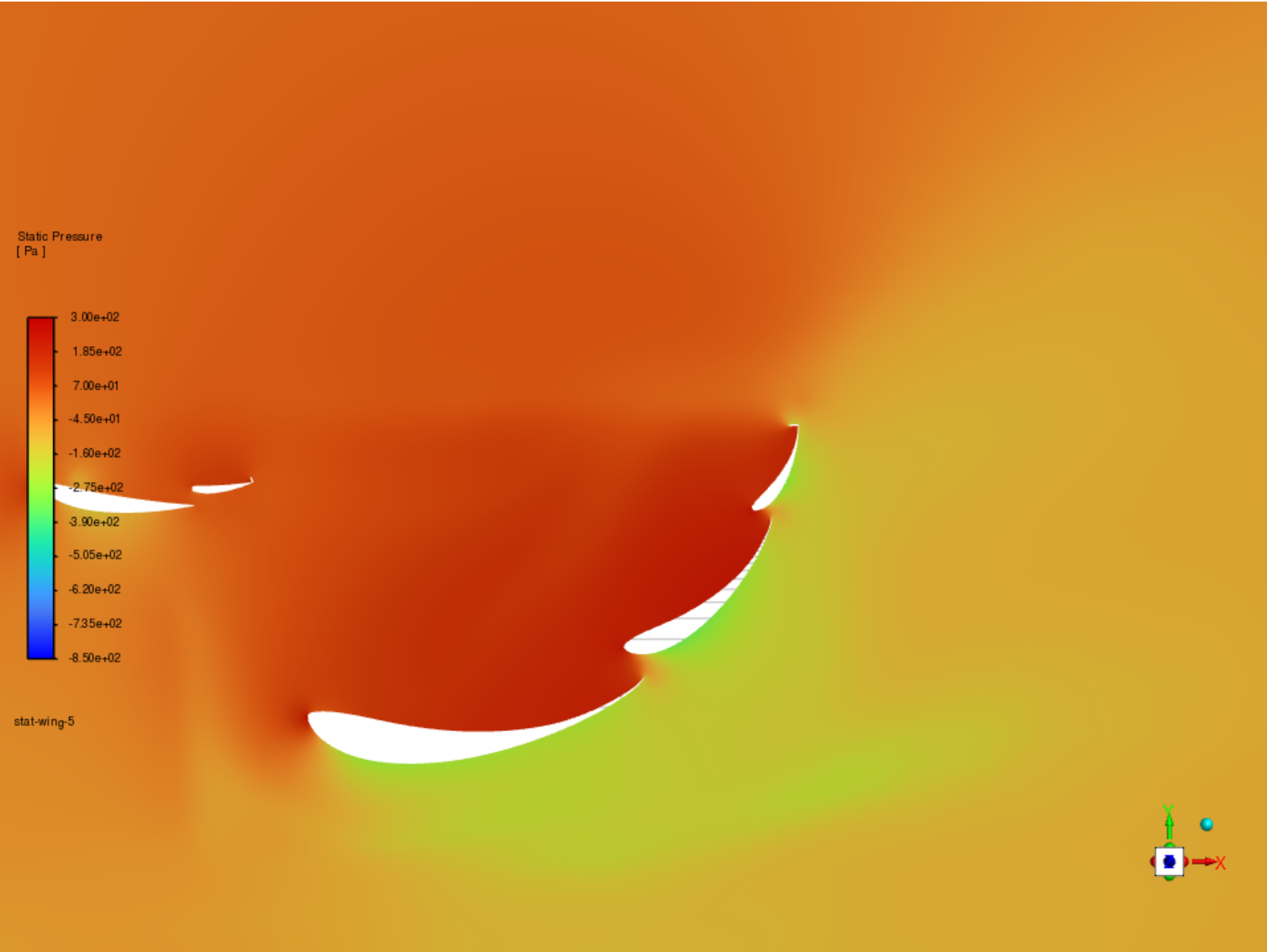
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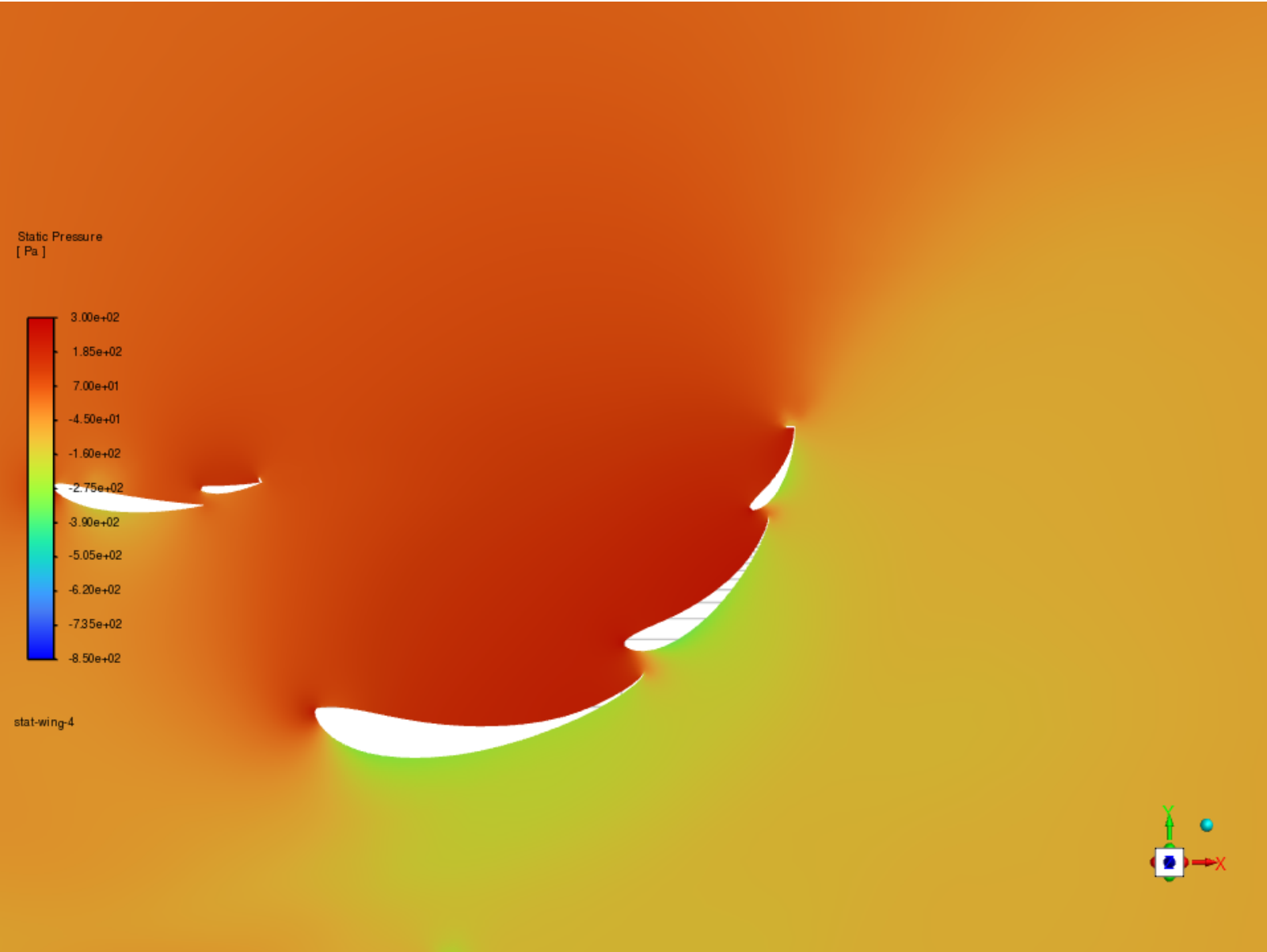
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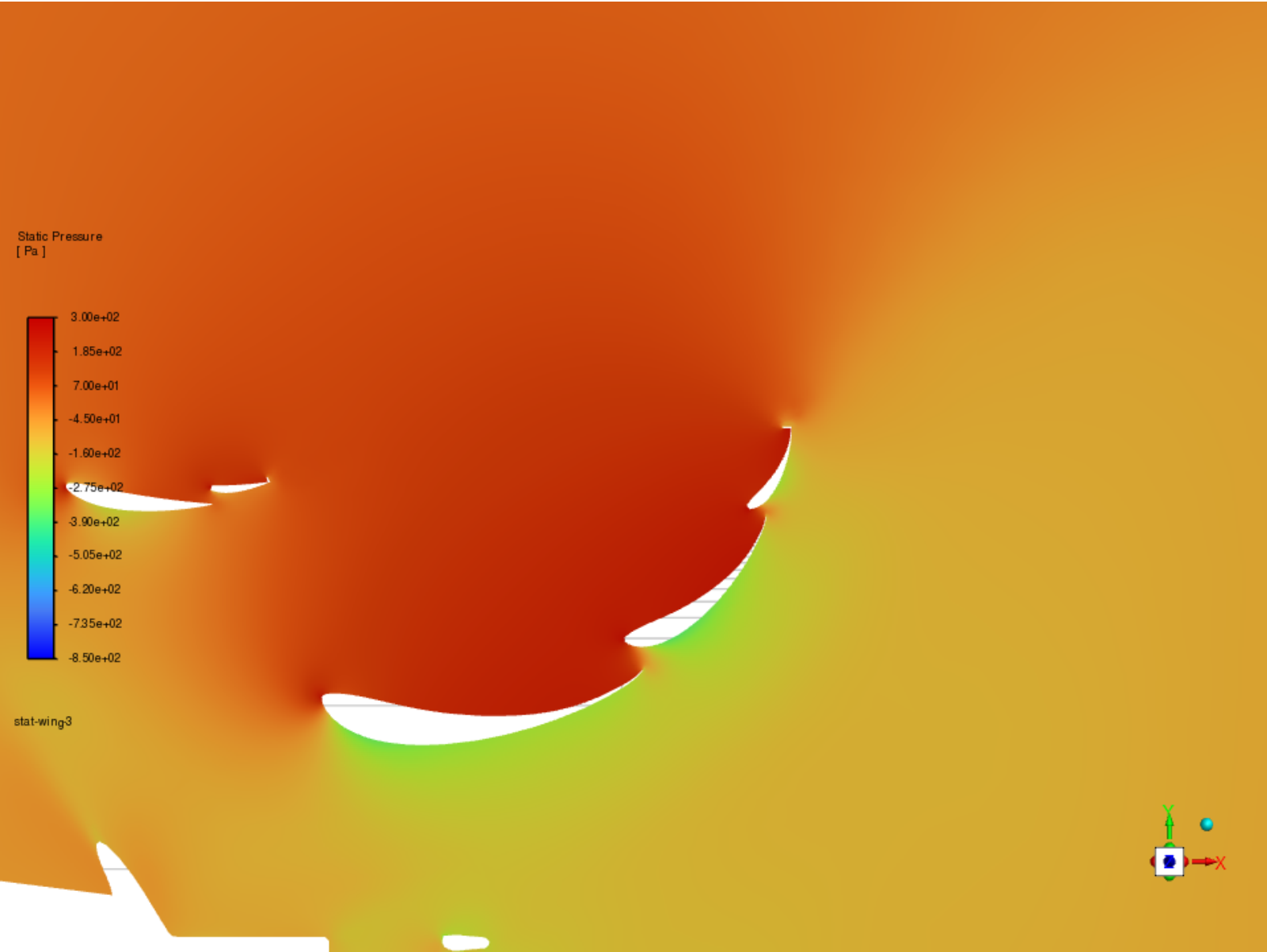
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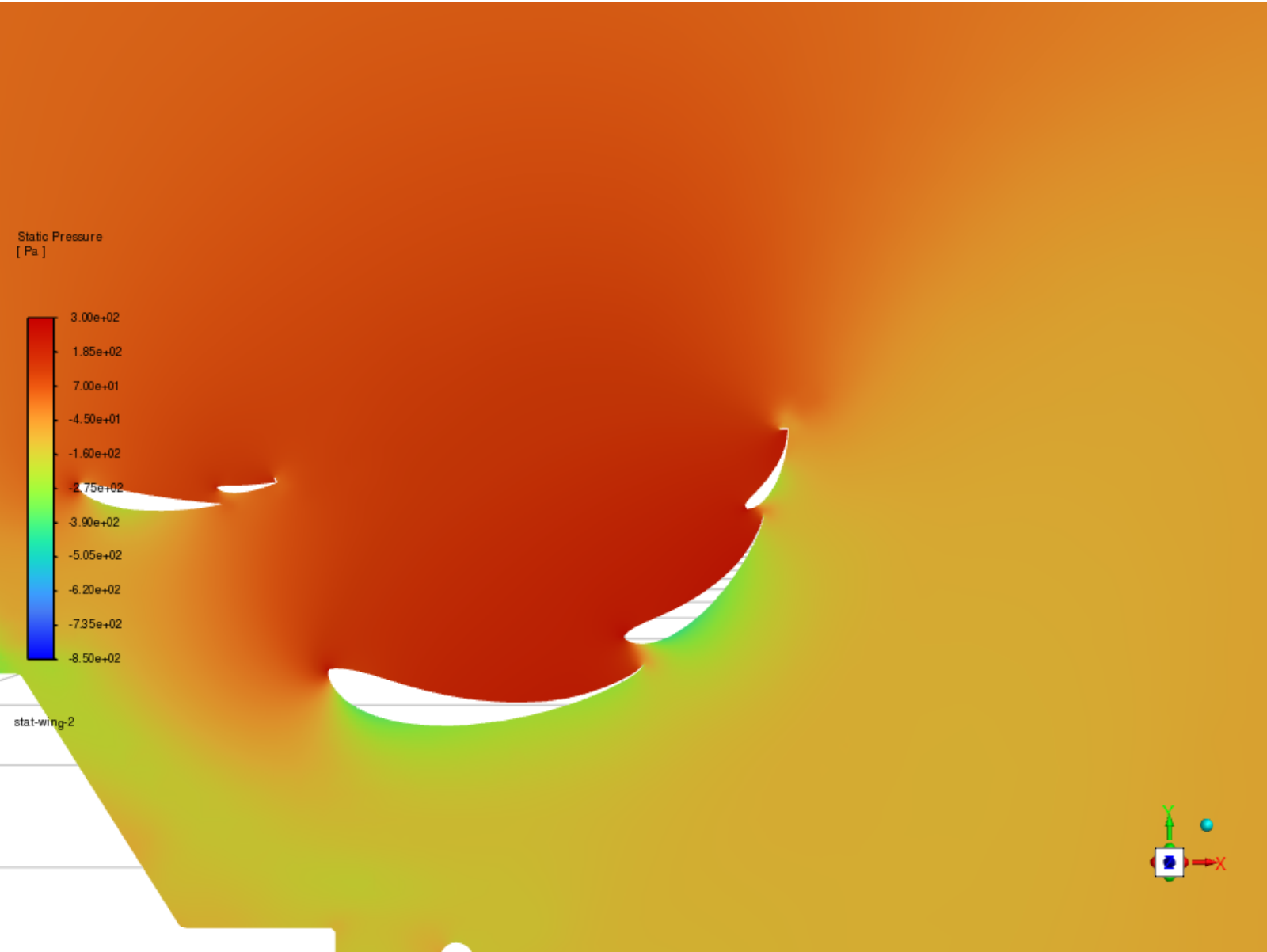
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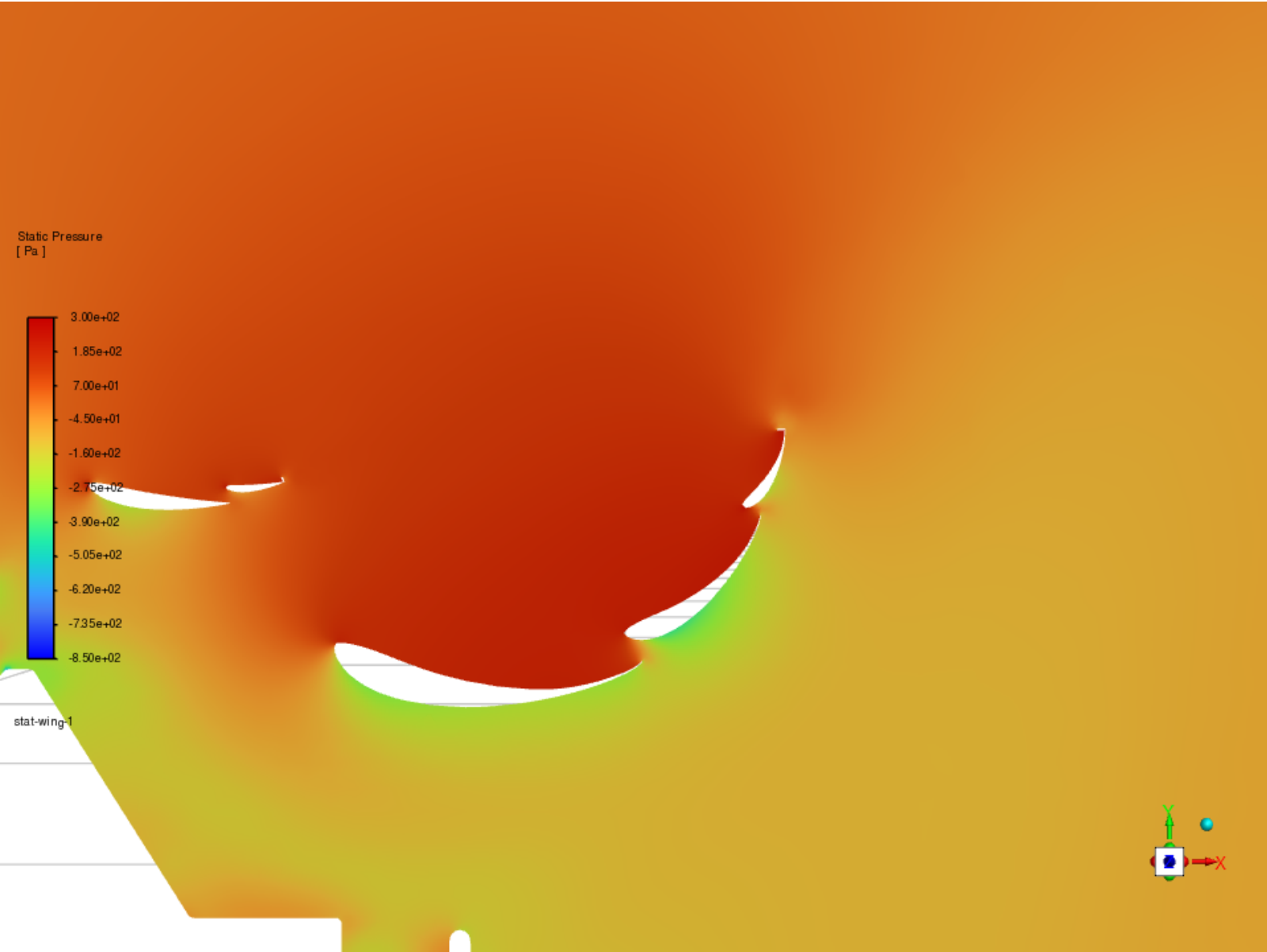
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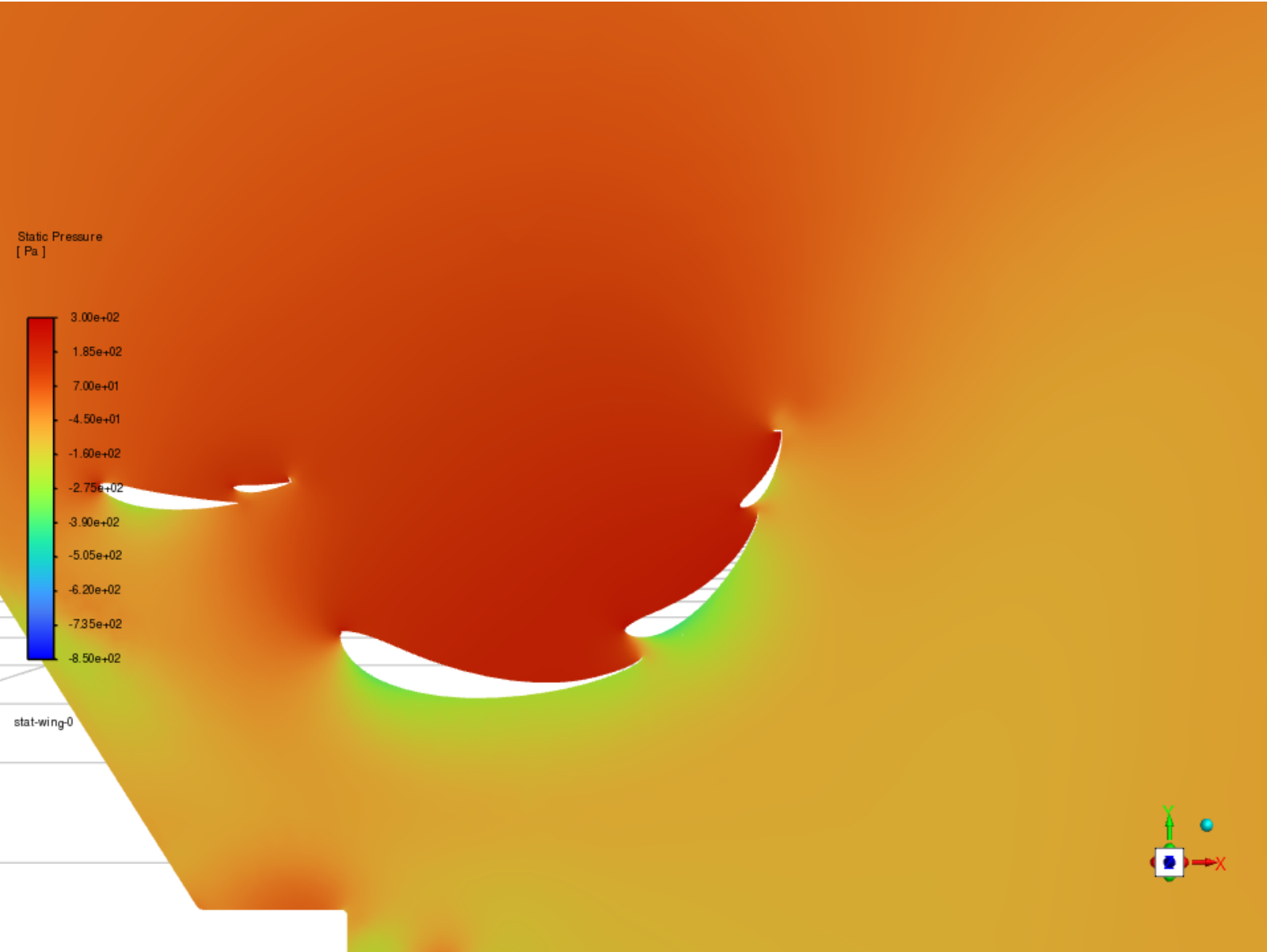
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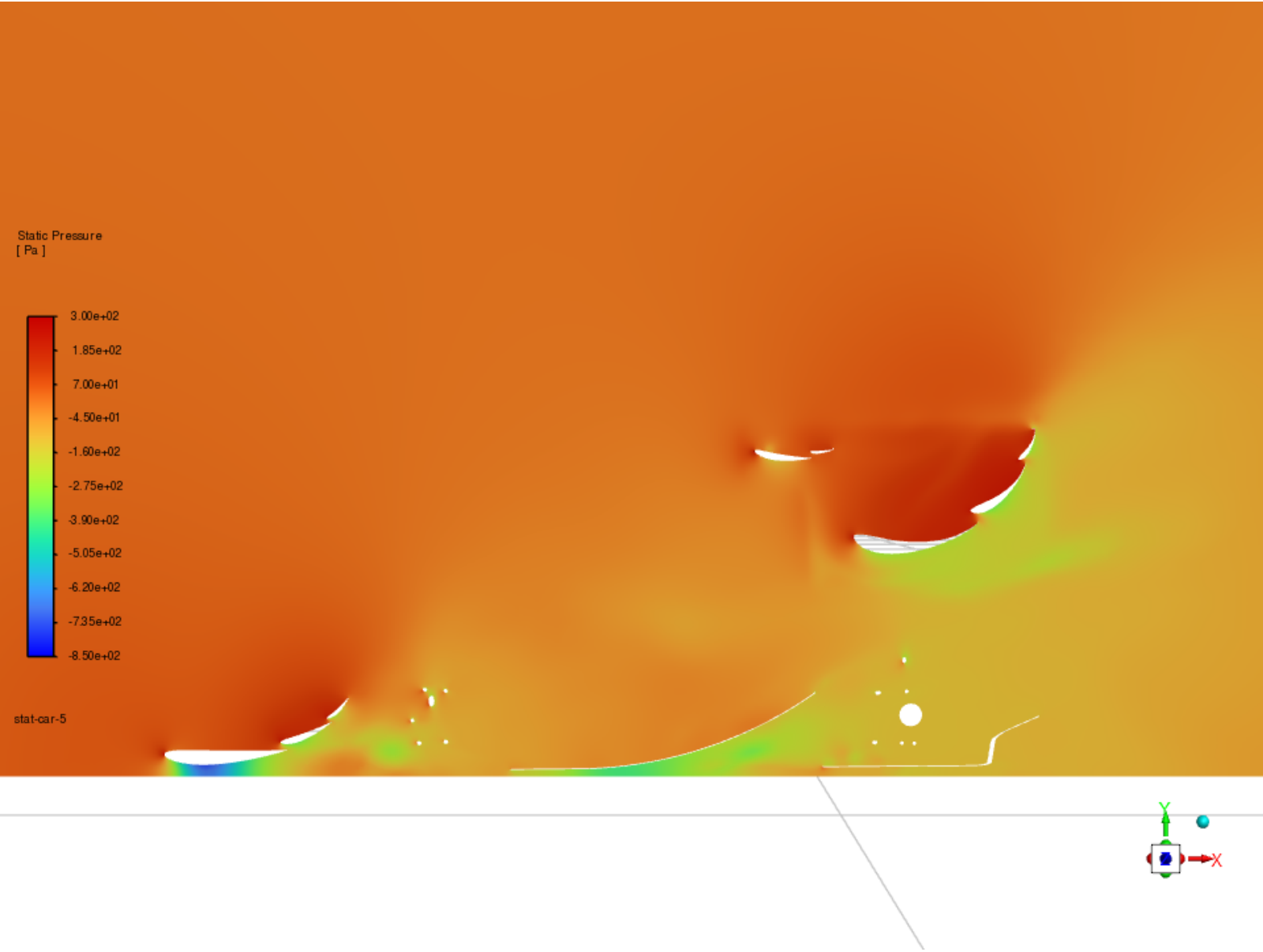
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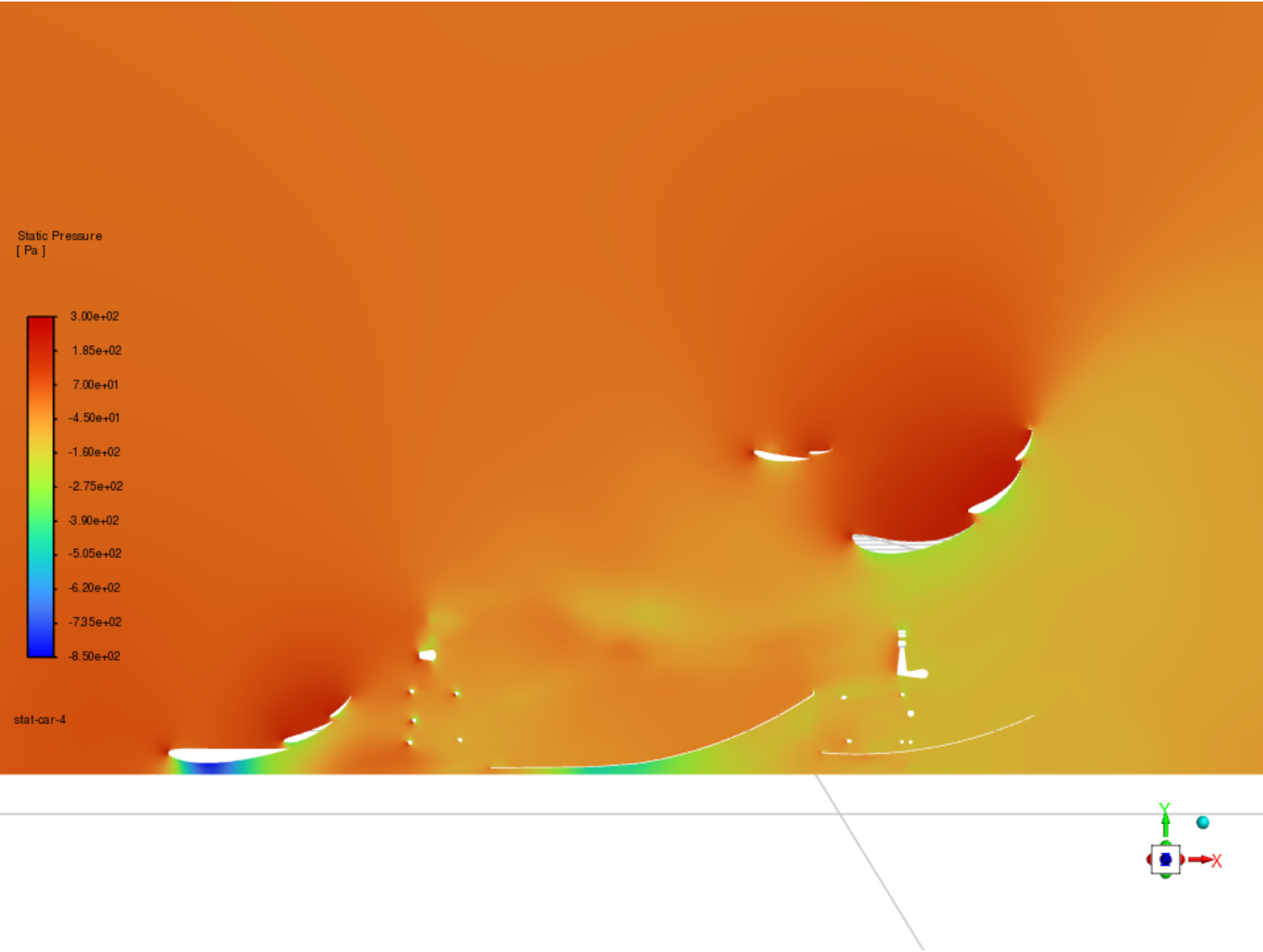
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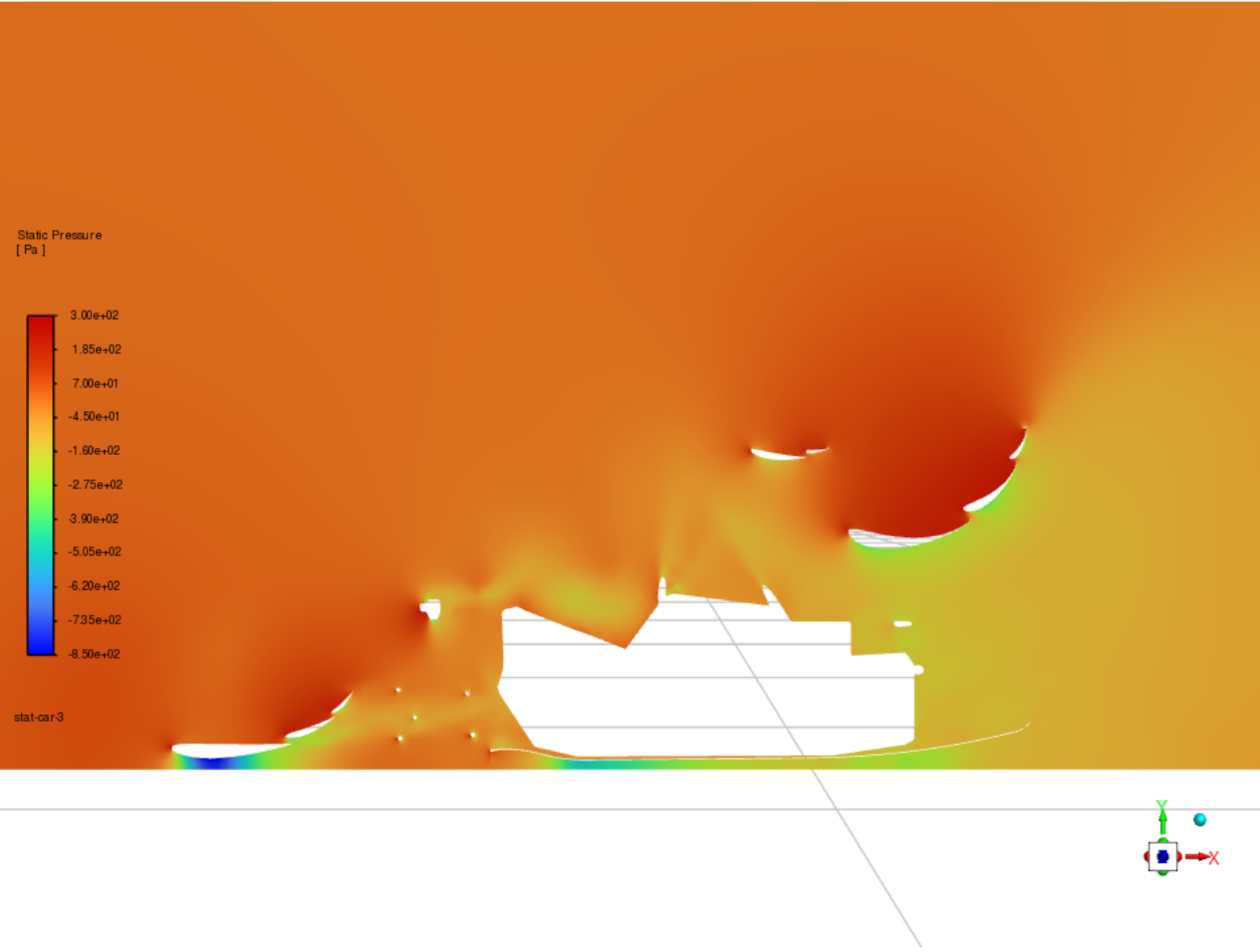
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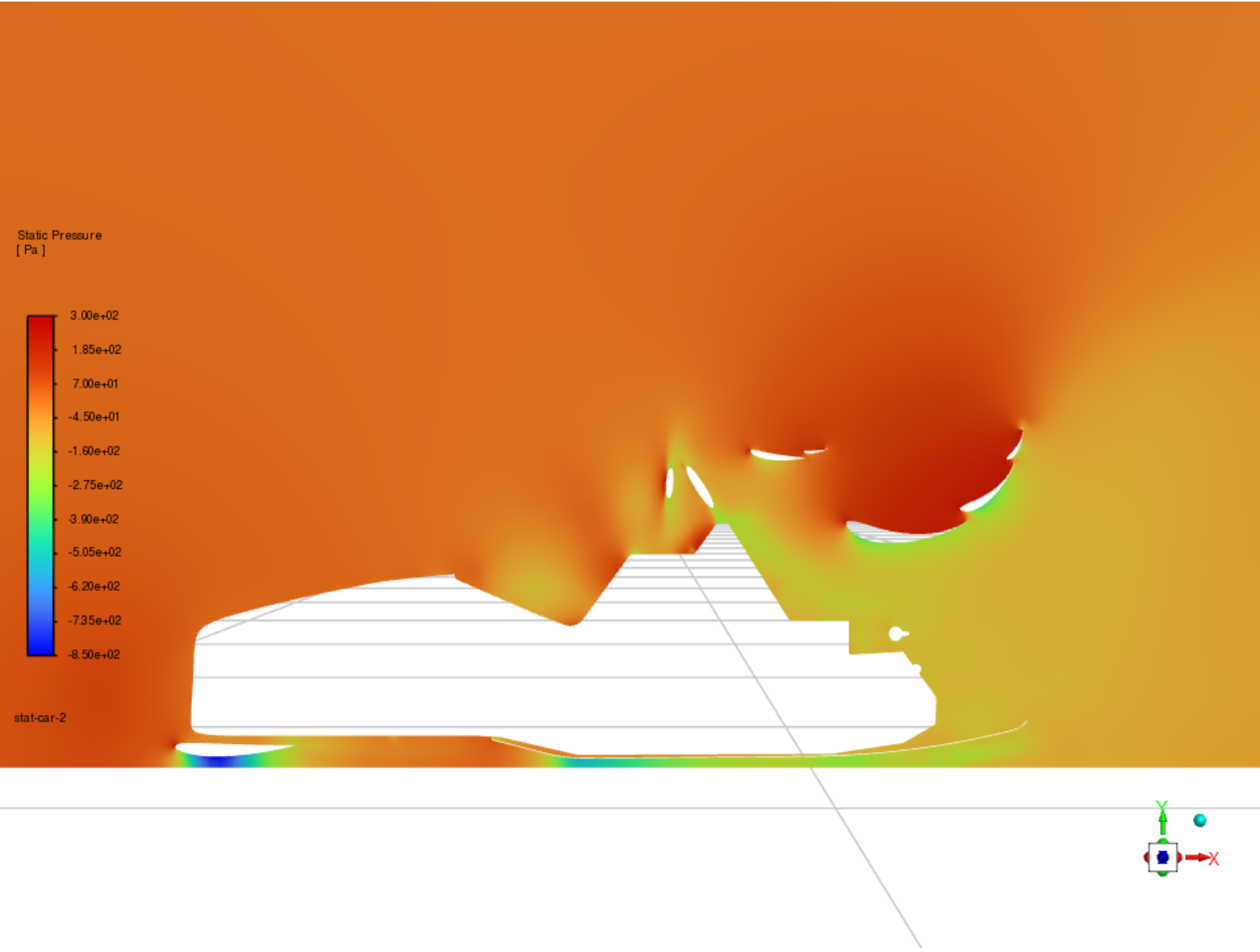
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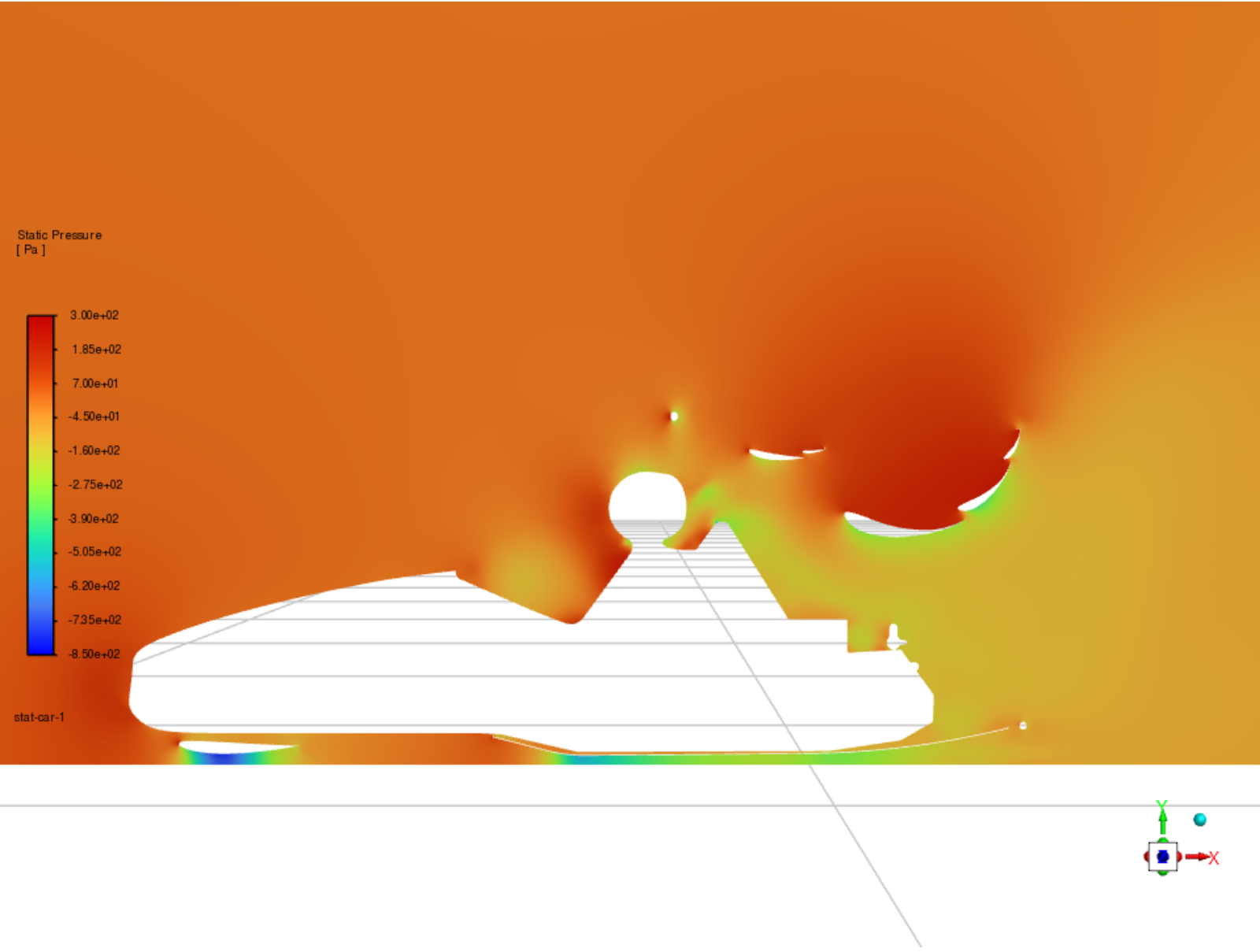
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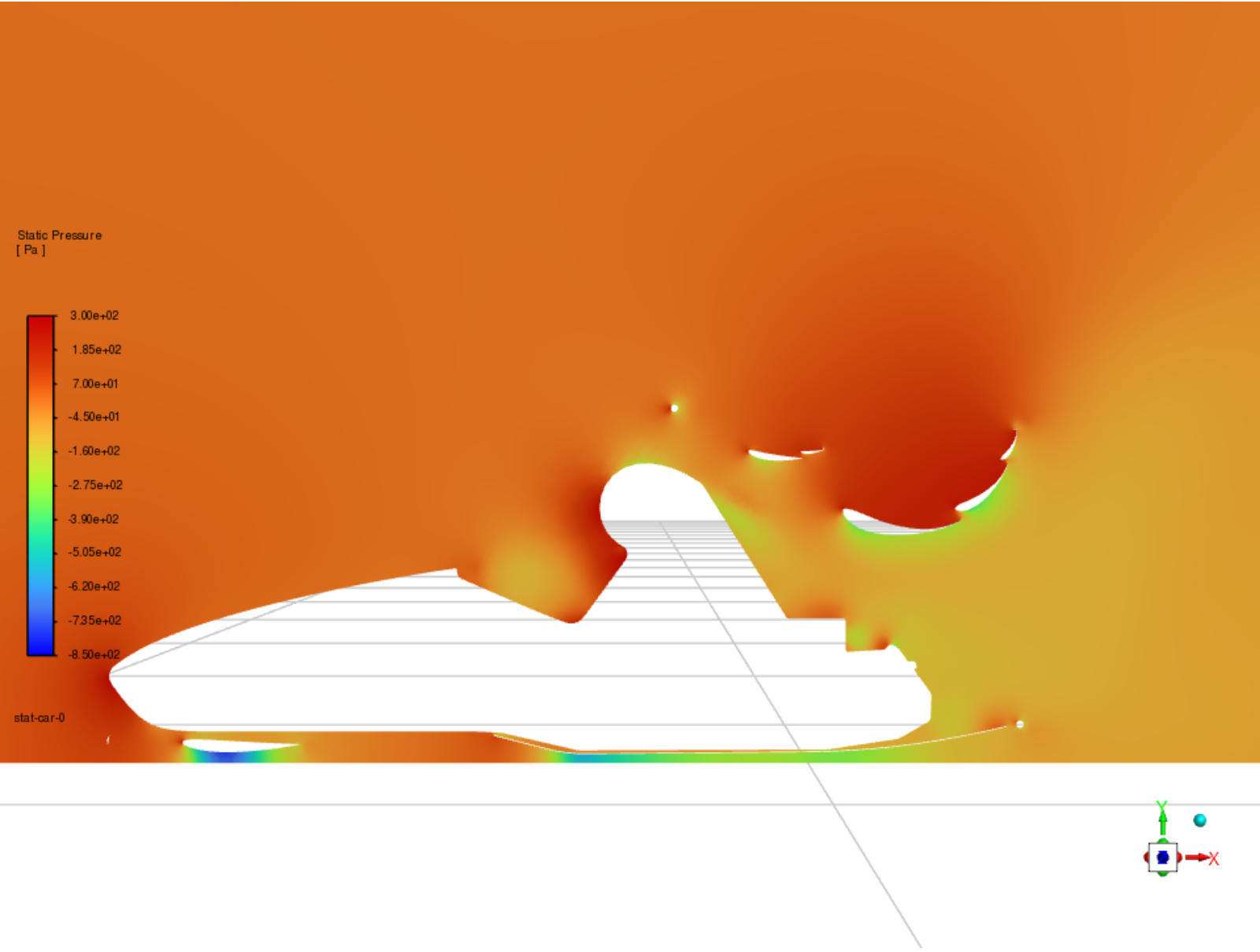
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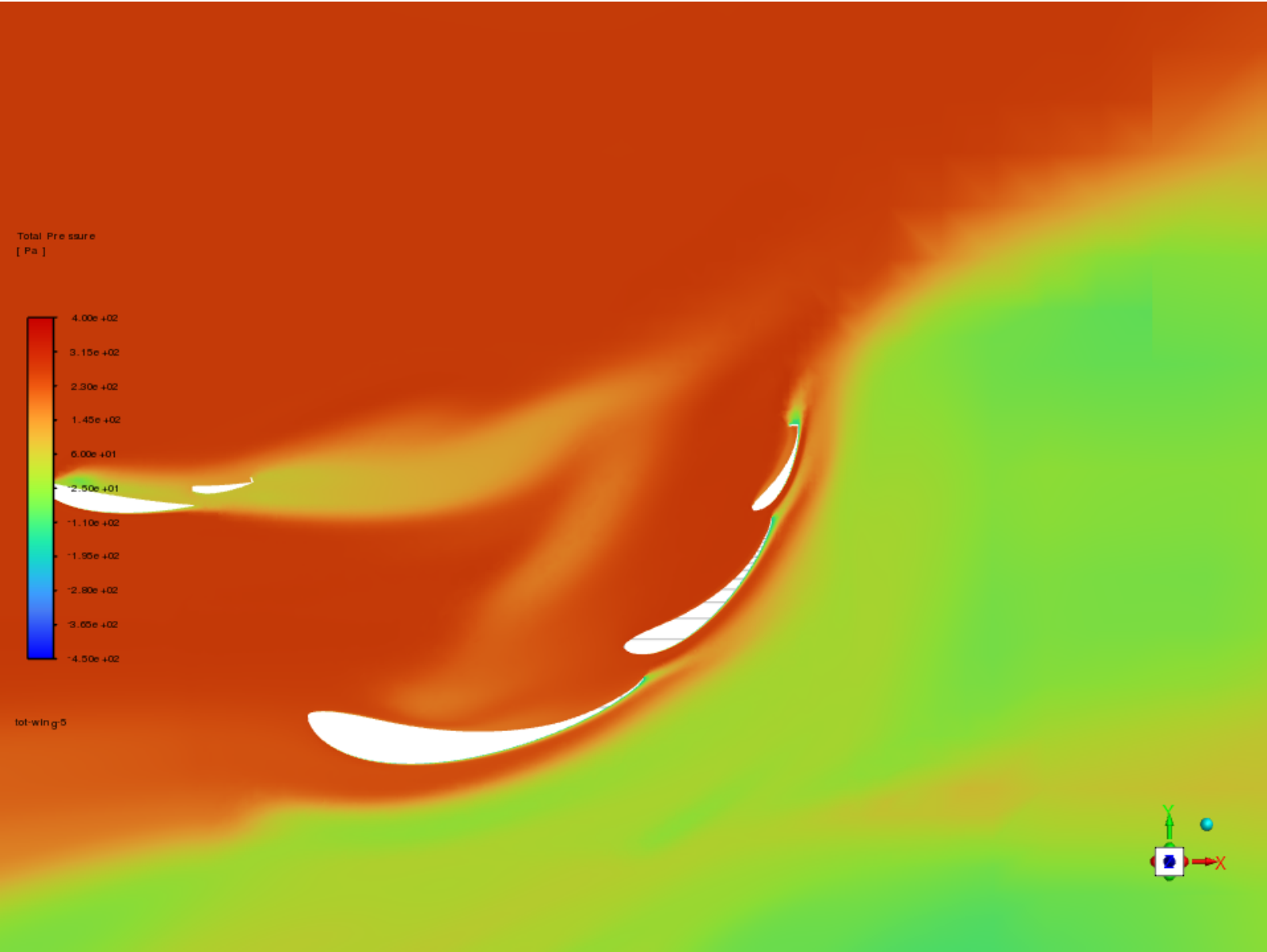
stat-car-1



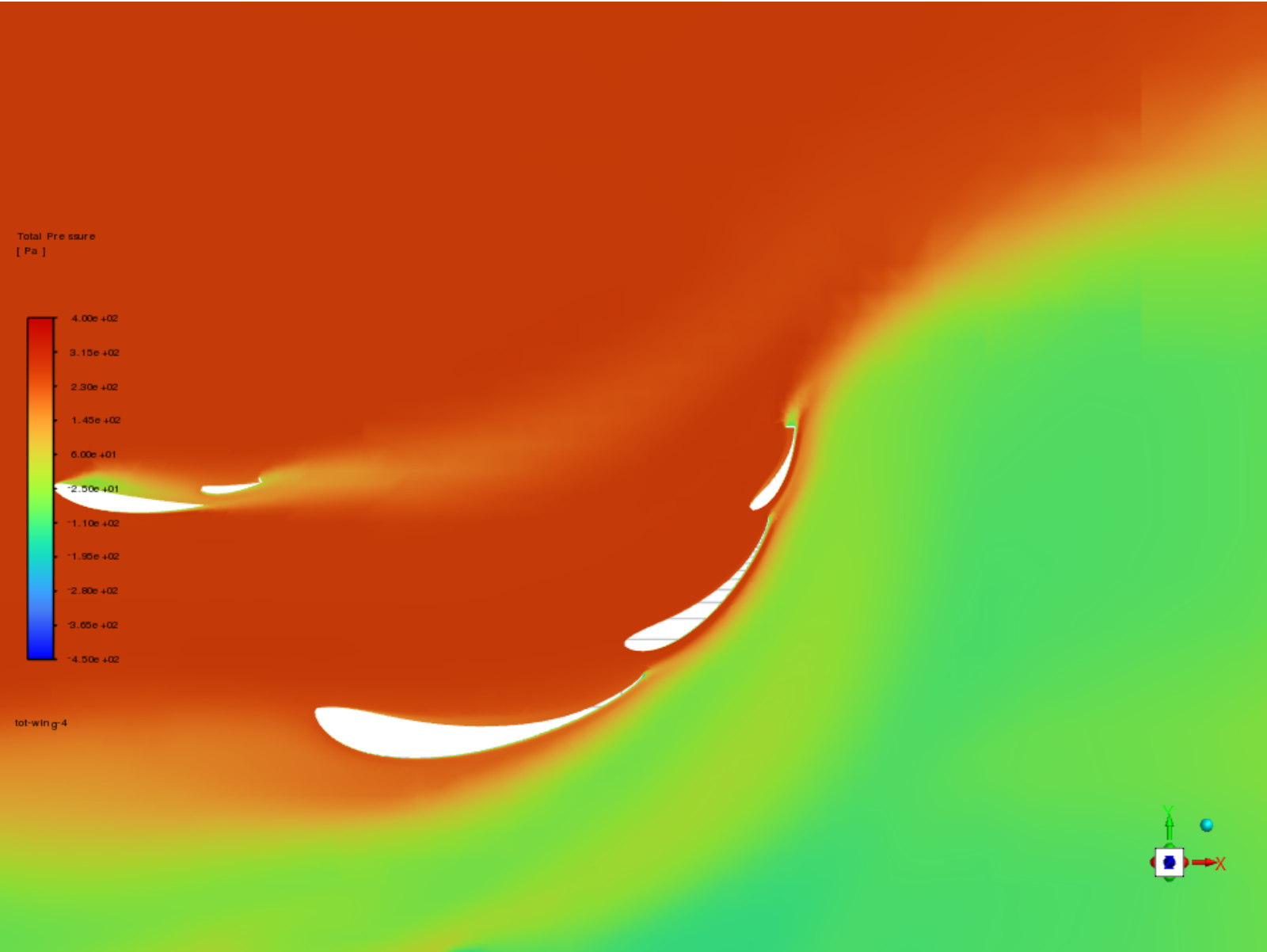
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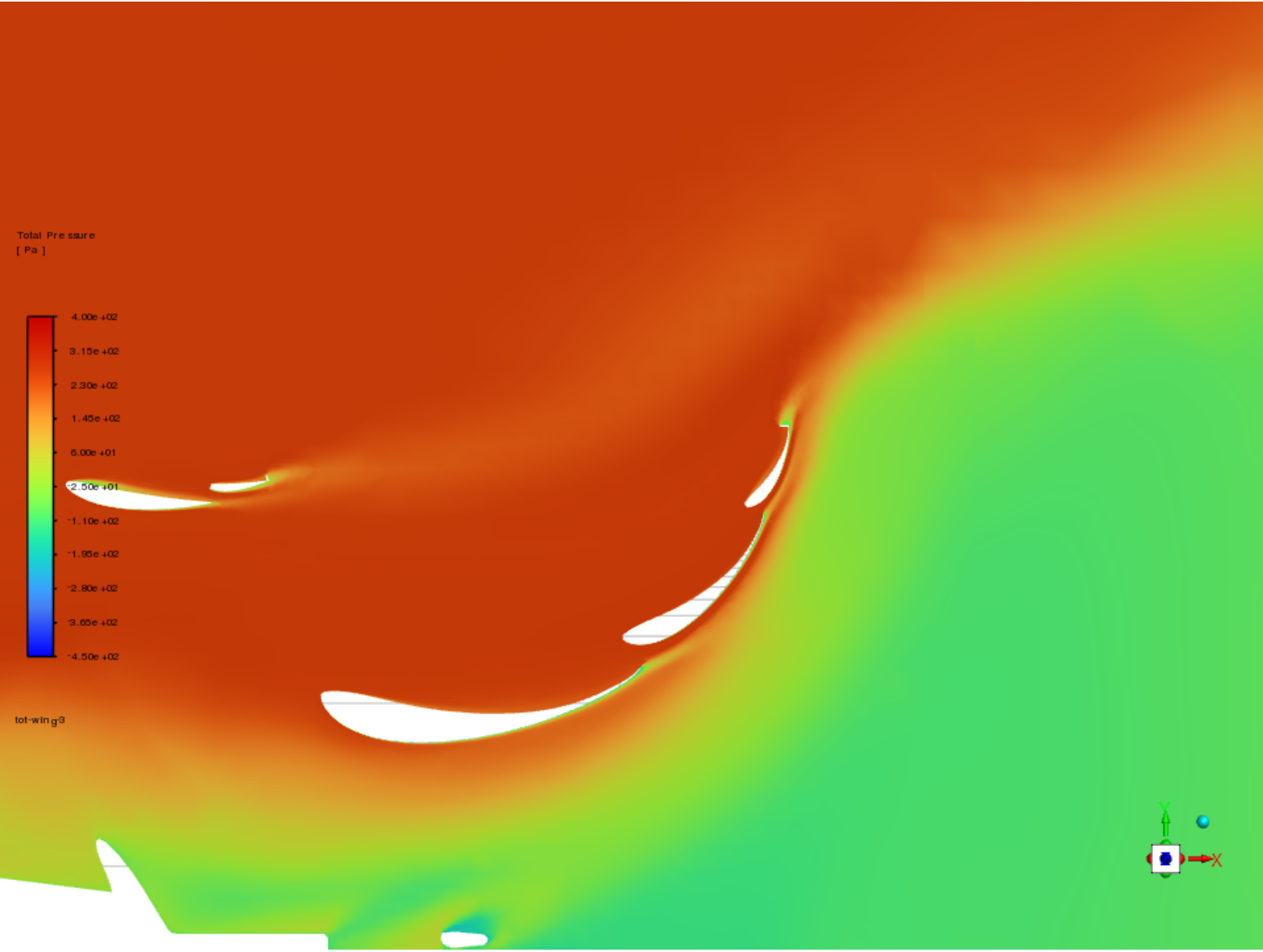
tot-wing-5



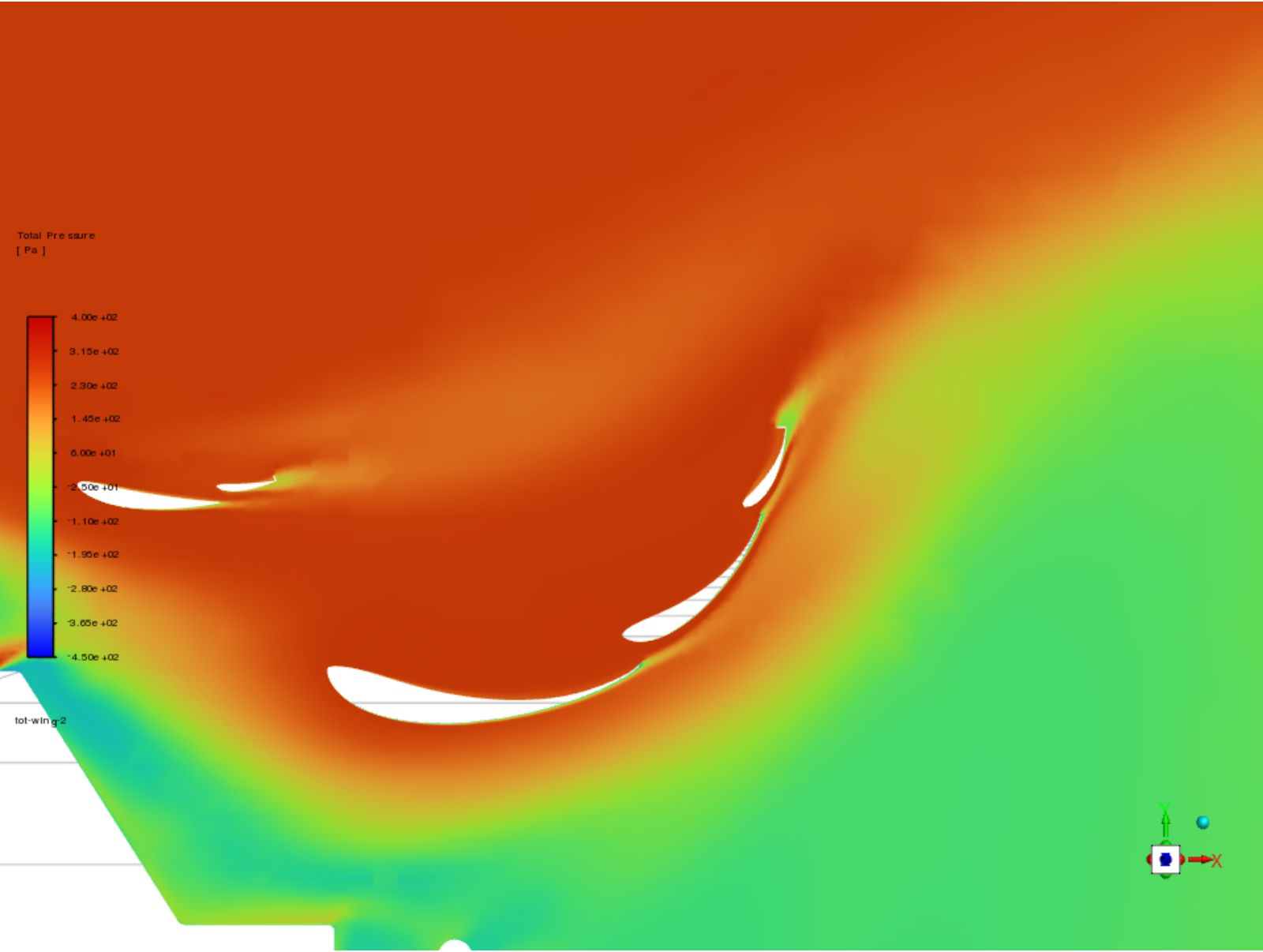
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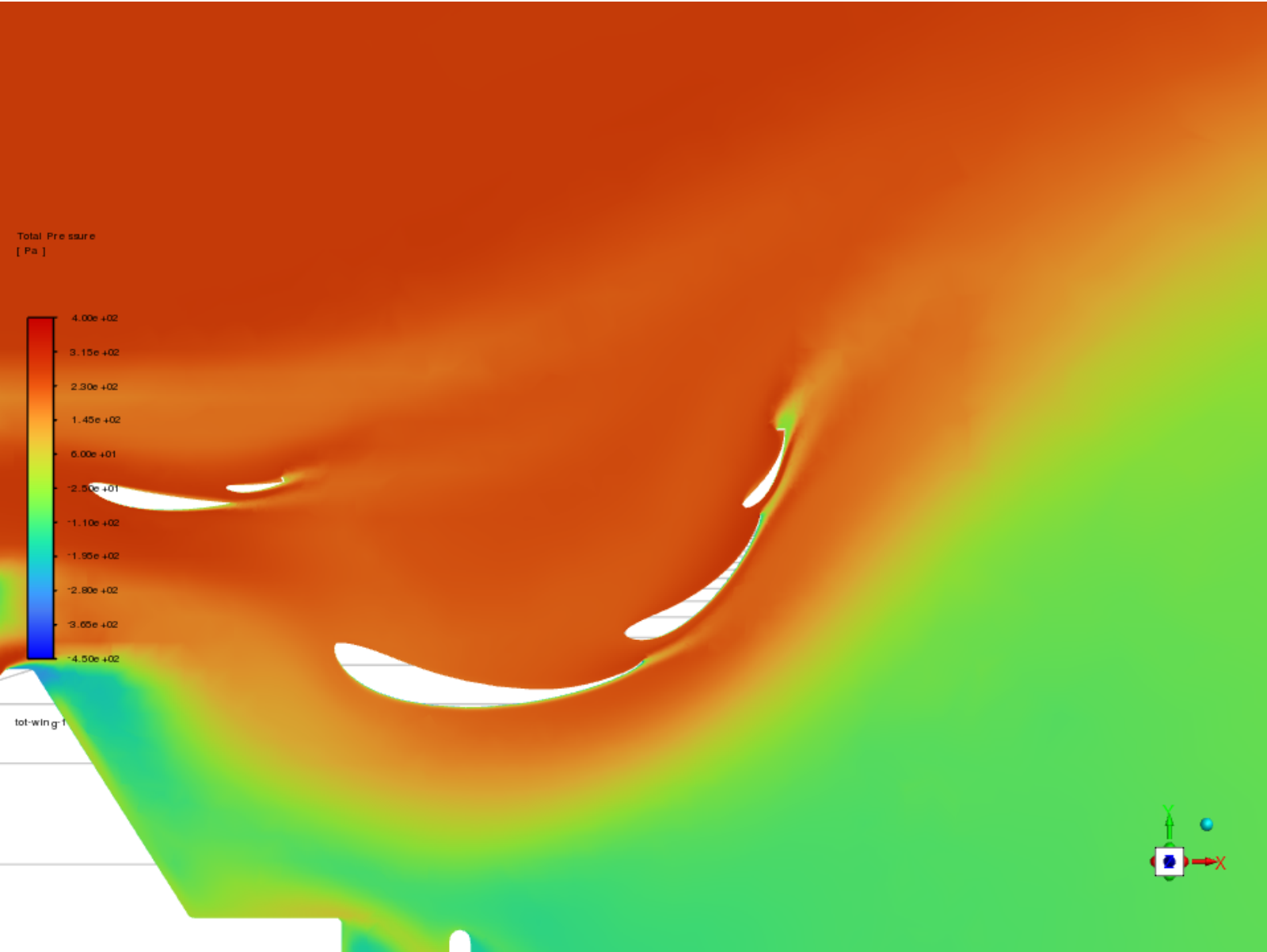
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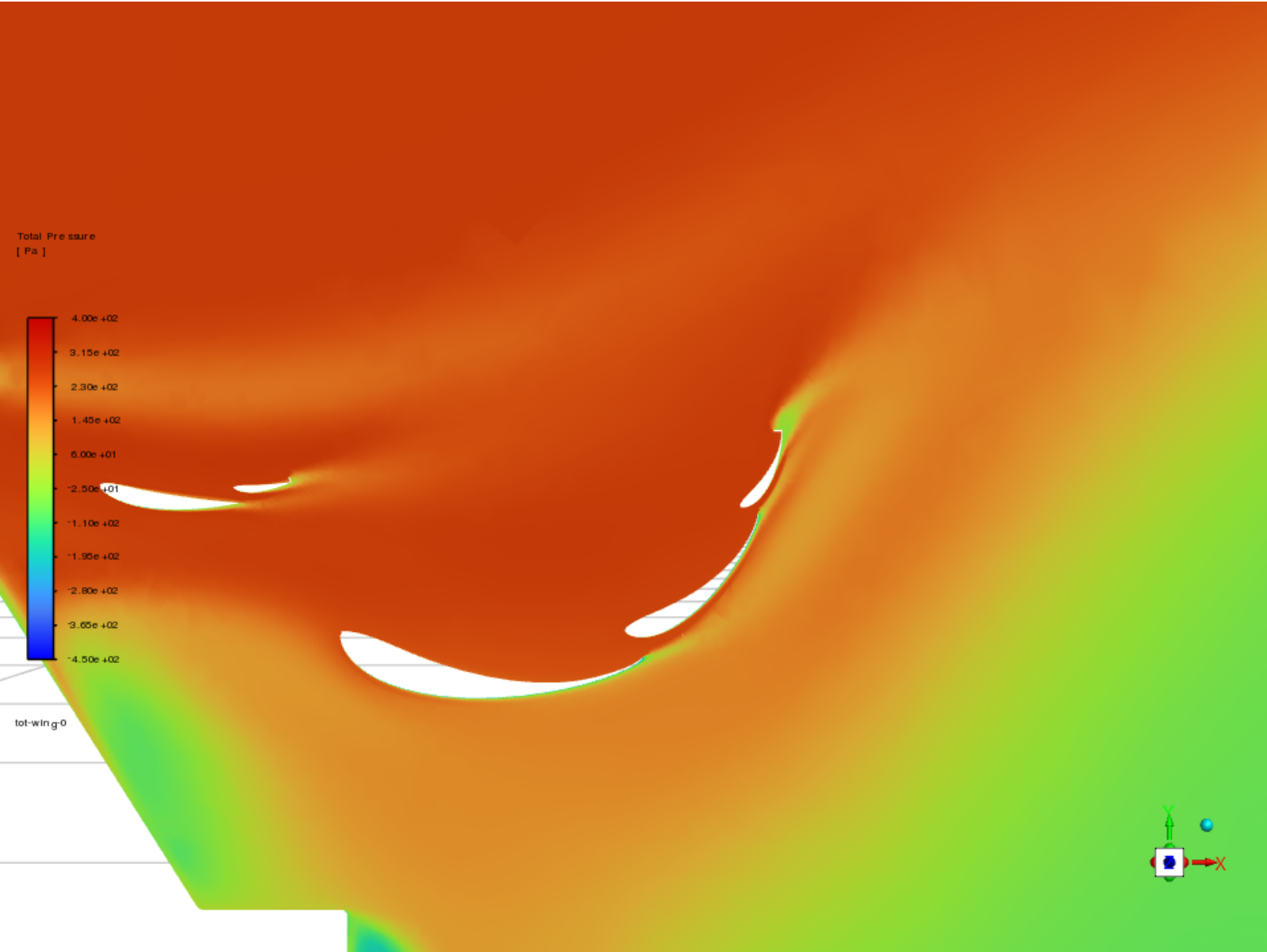
tot-wing-2



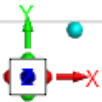
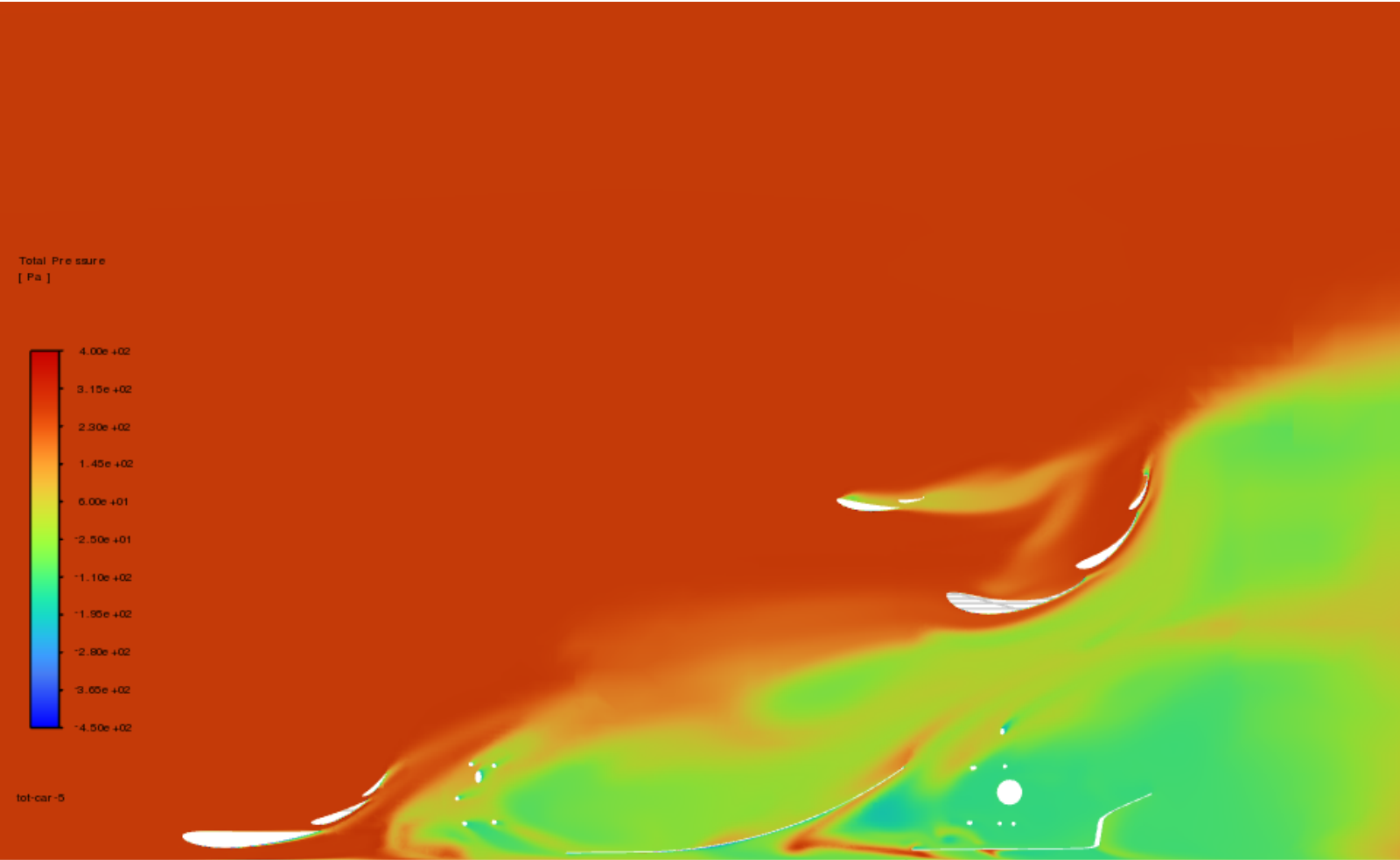
tot-wing-1



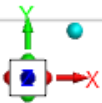
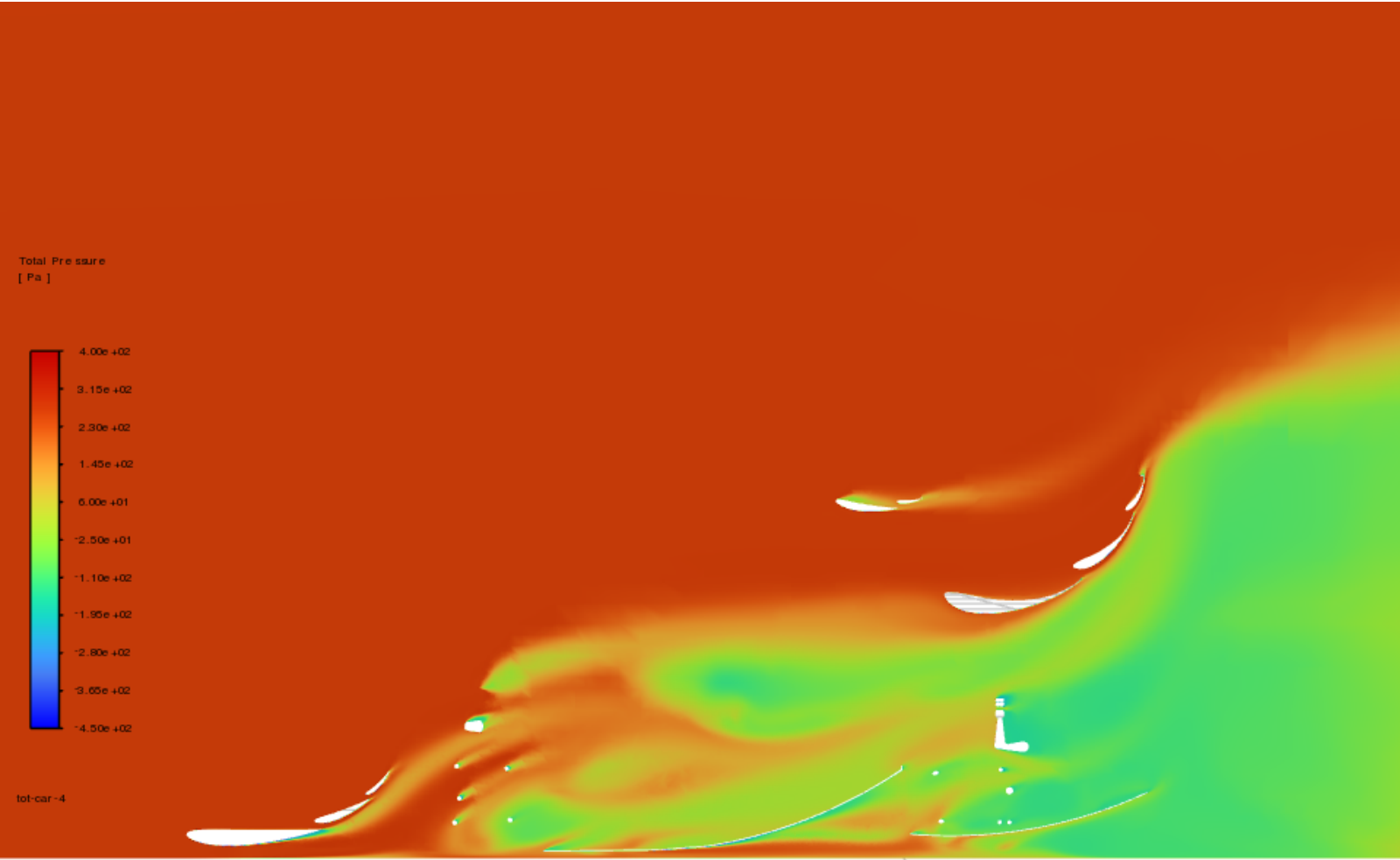
tot-wing-0



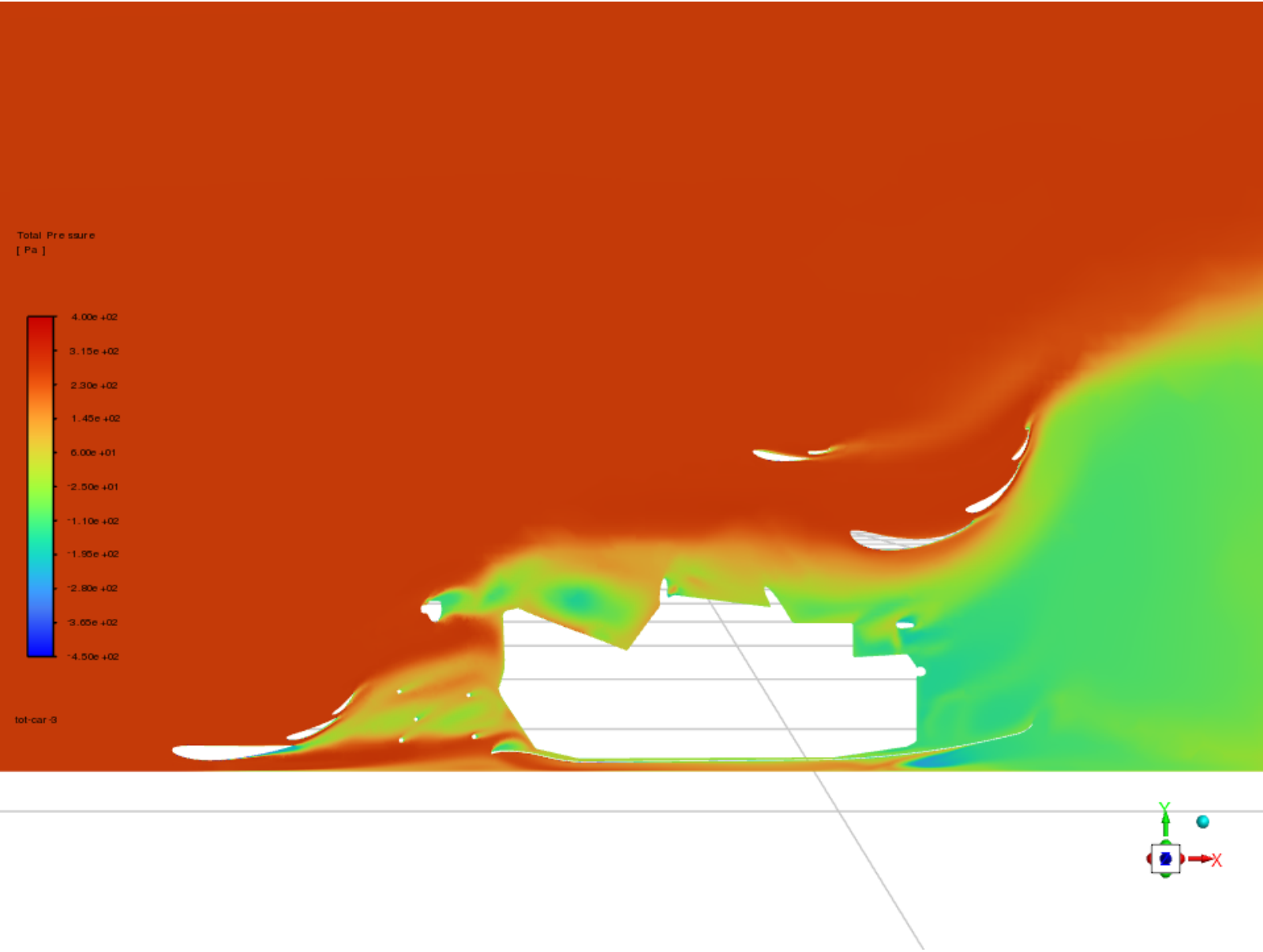
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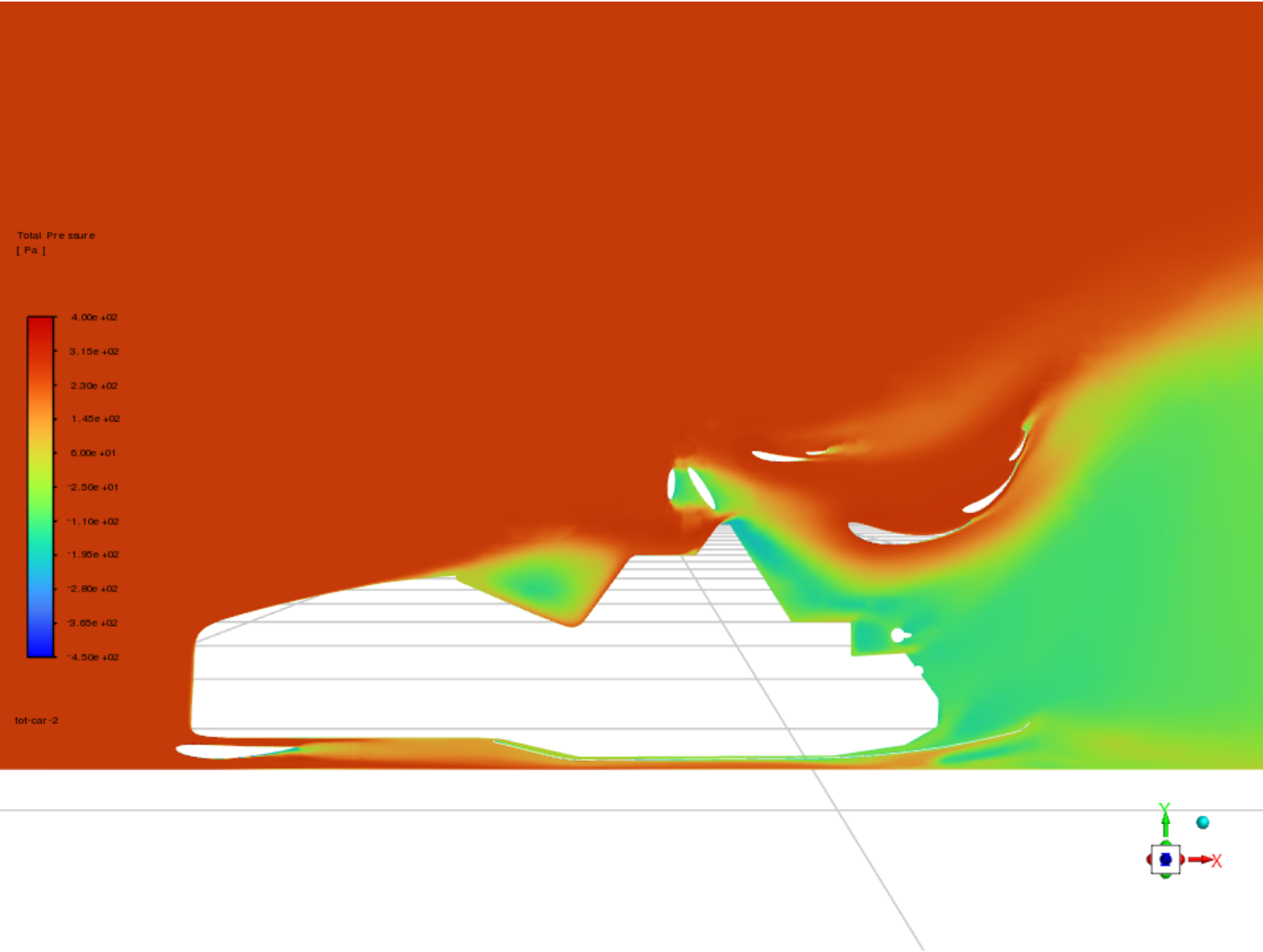
tot-car-4



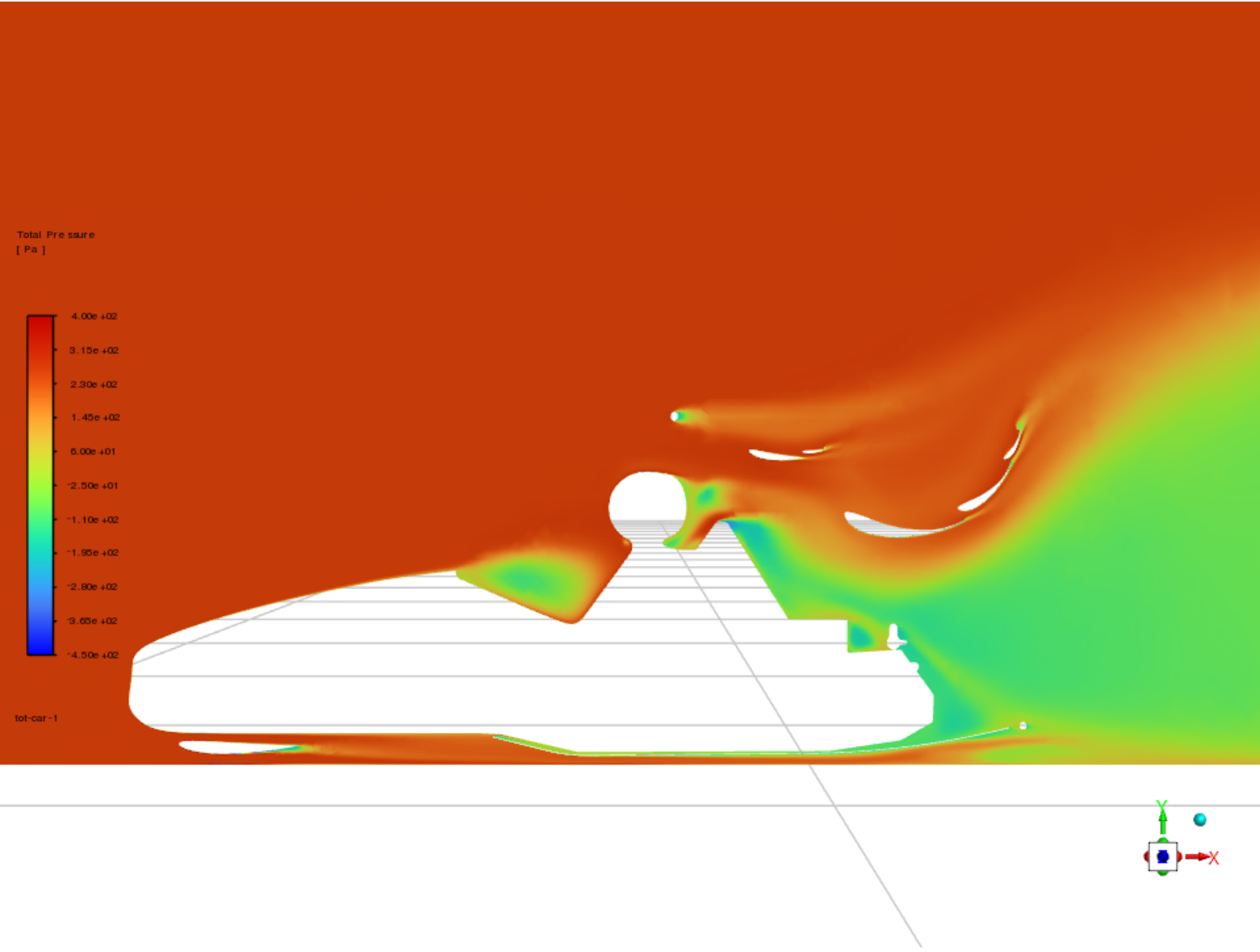
tot-car-3



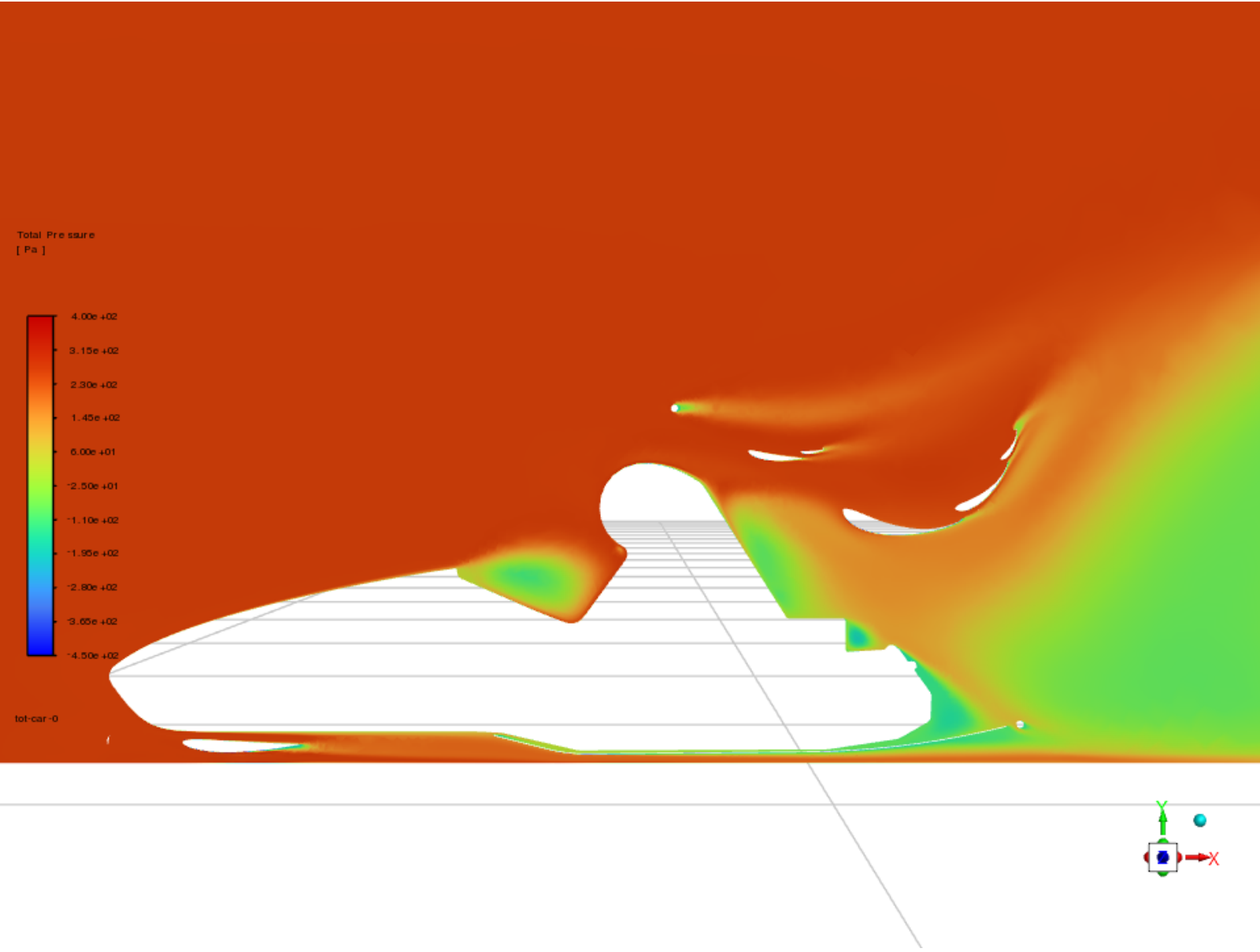
tot-car-2



tot-car-1

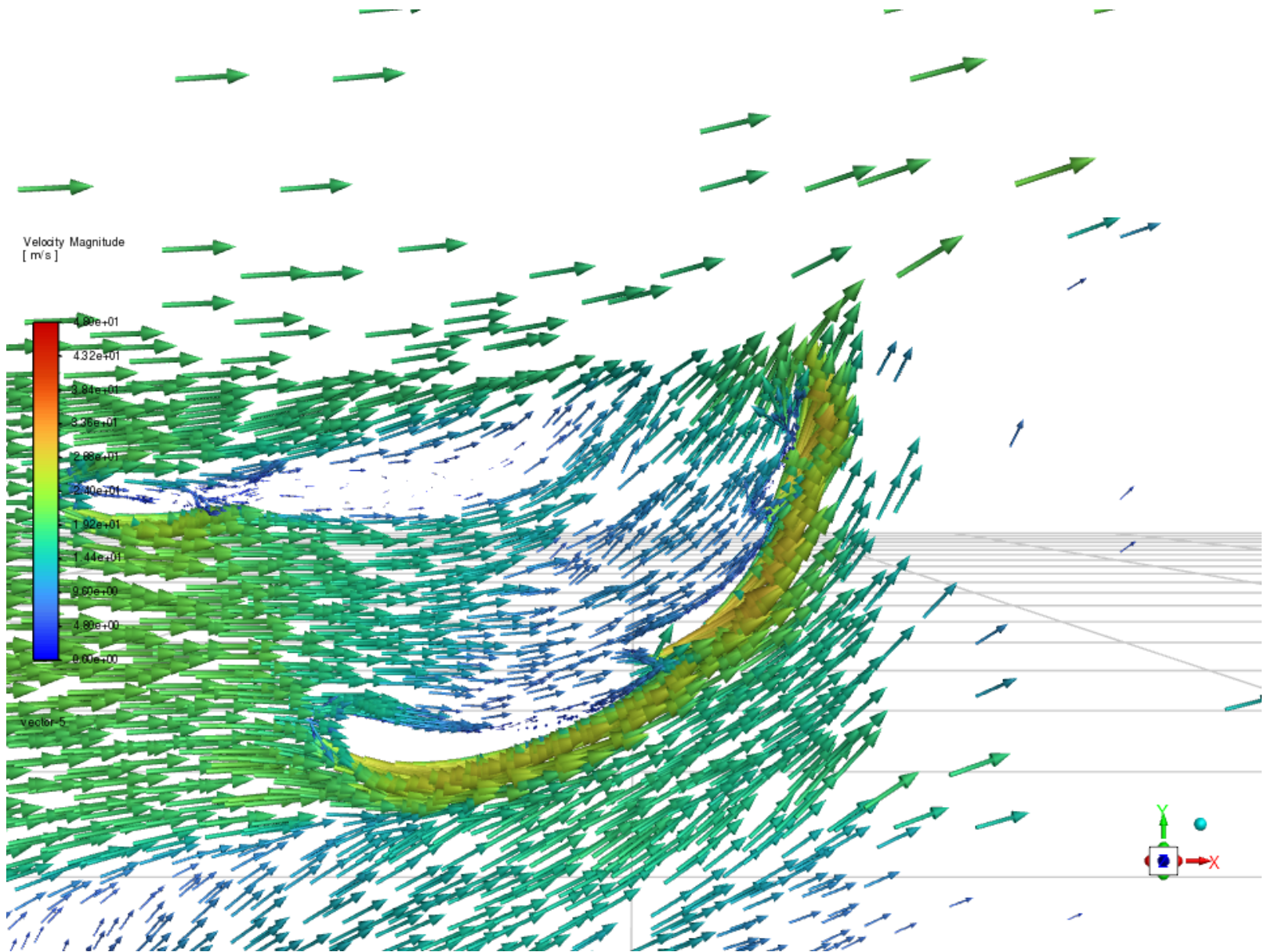


tot-car-0

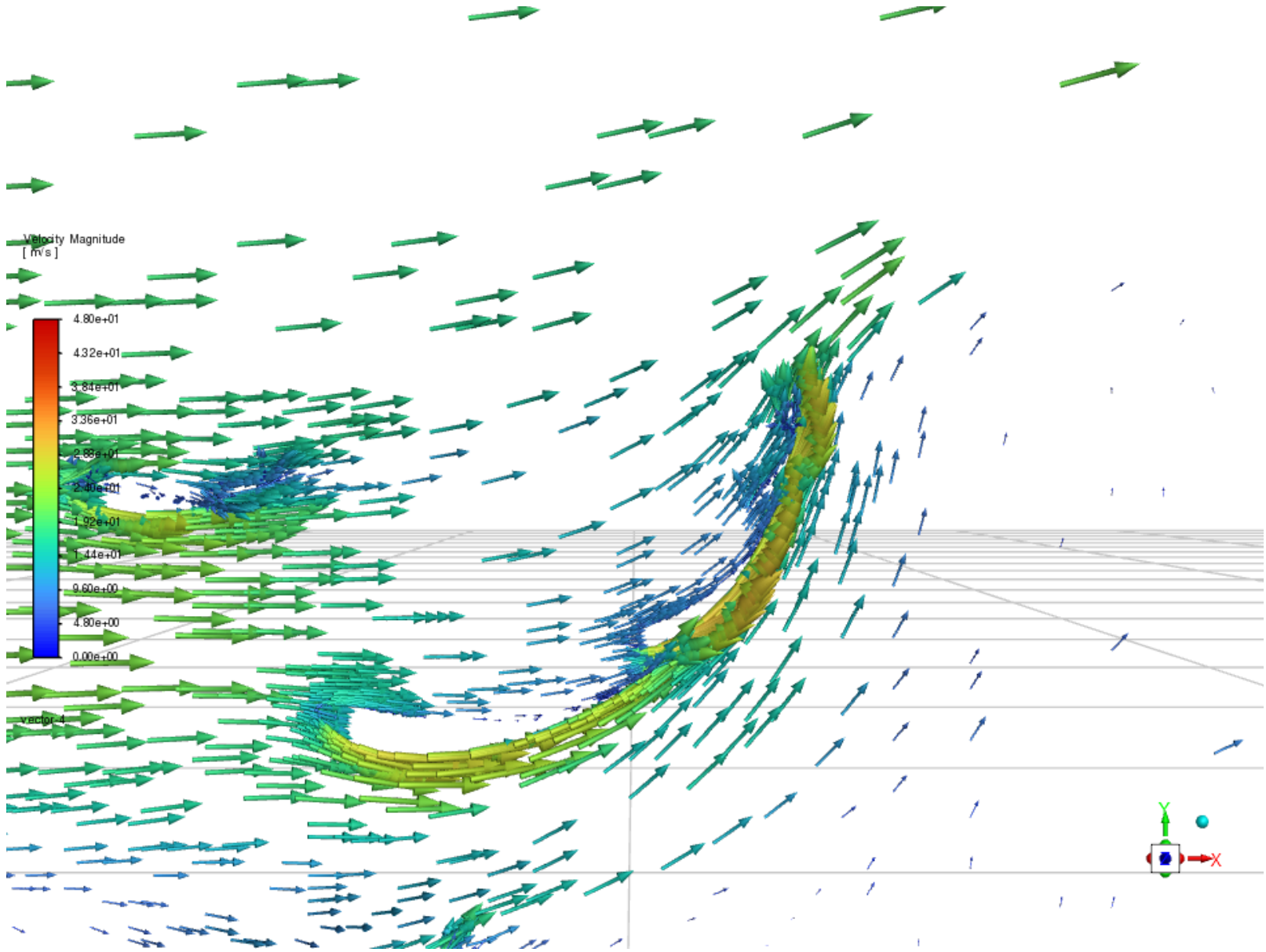


Vectors

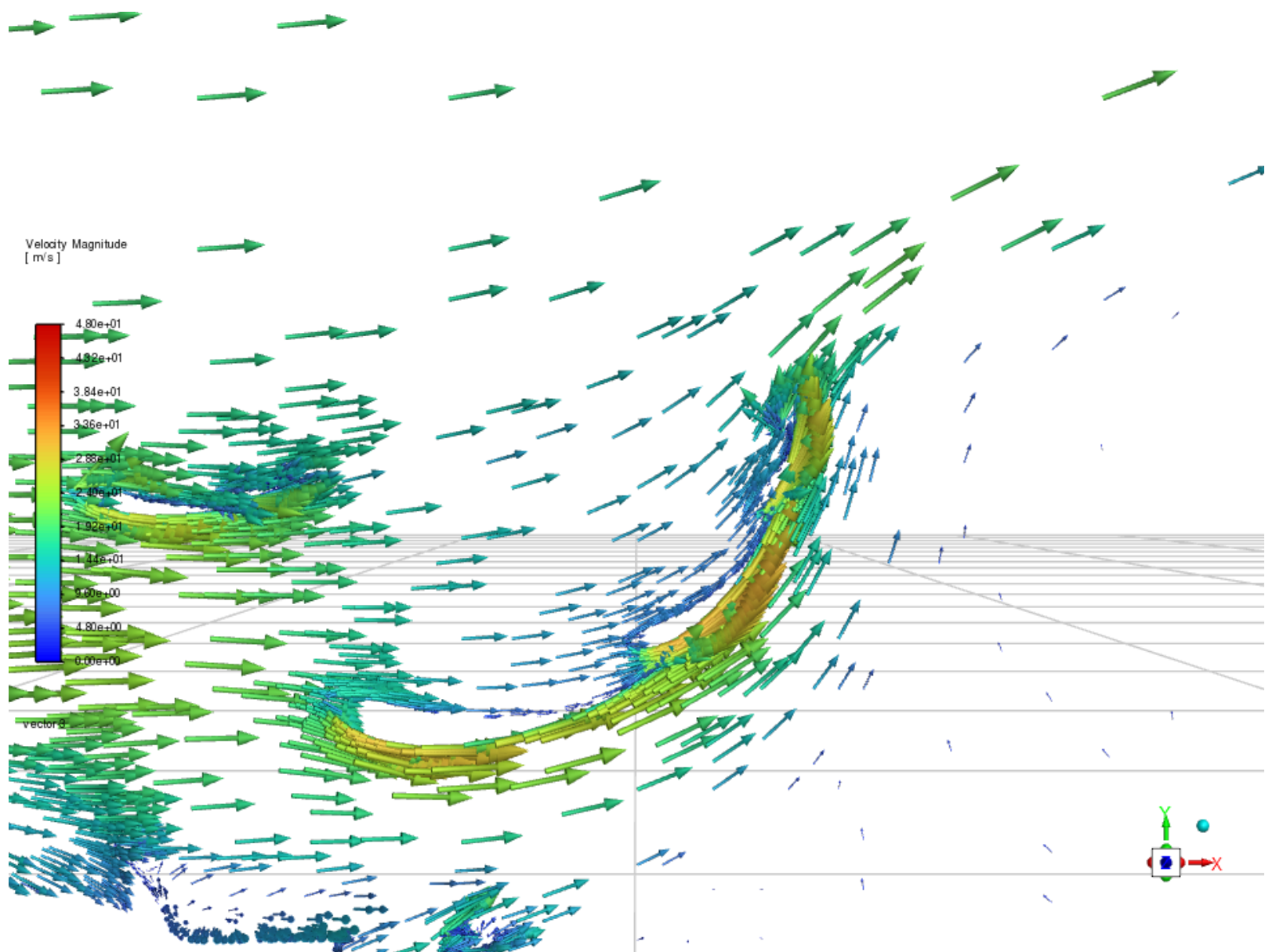
vector-5



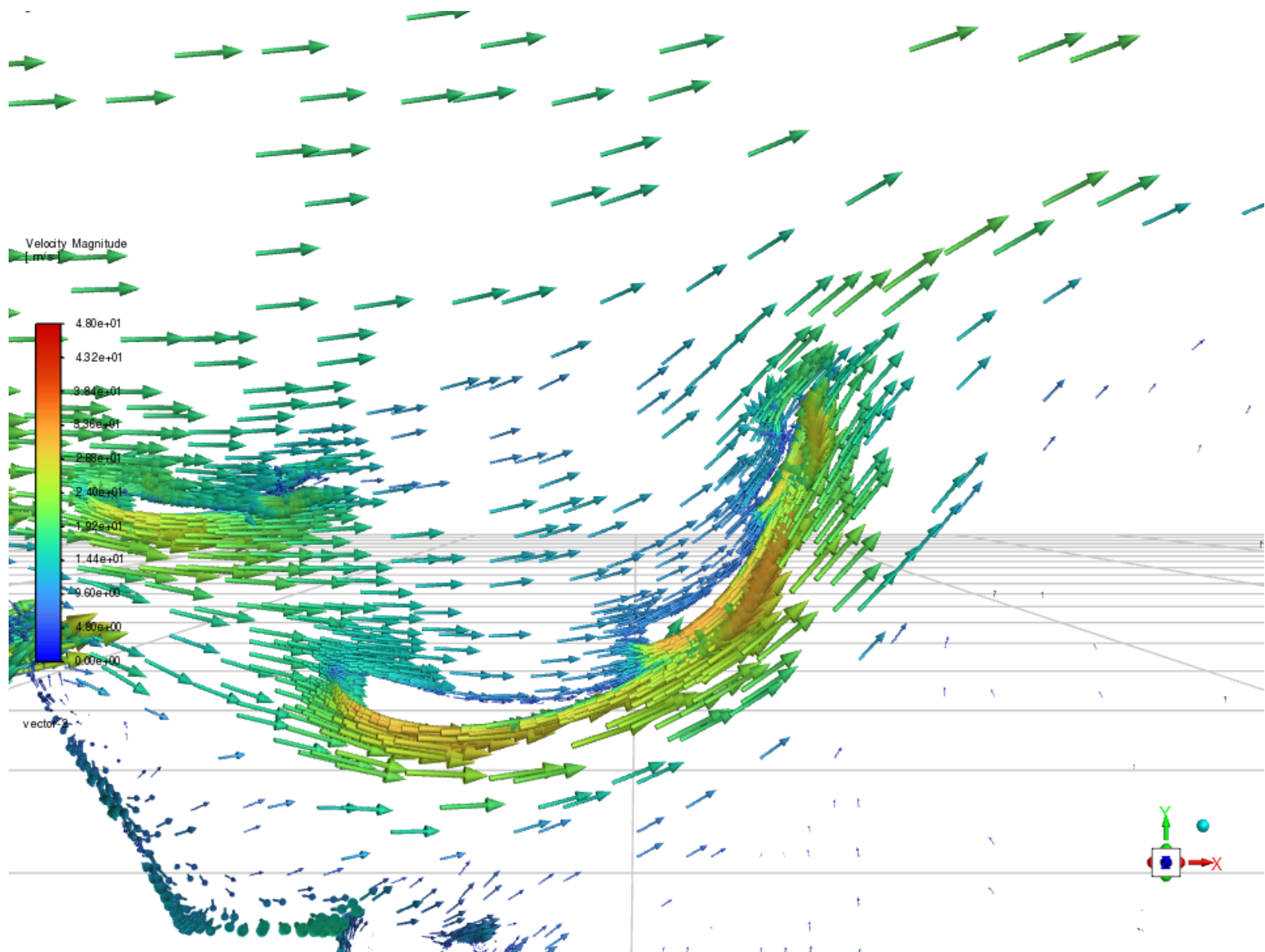
vector-4



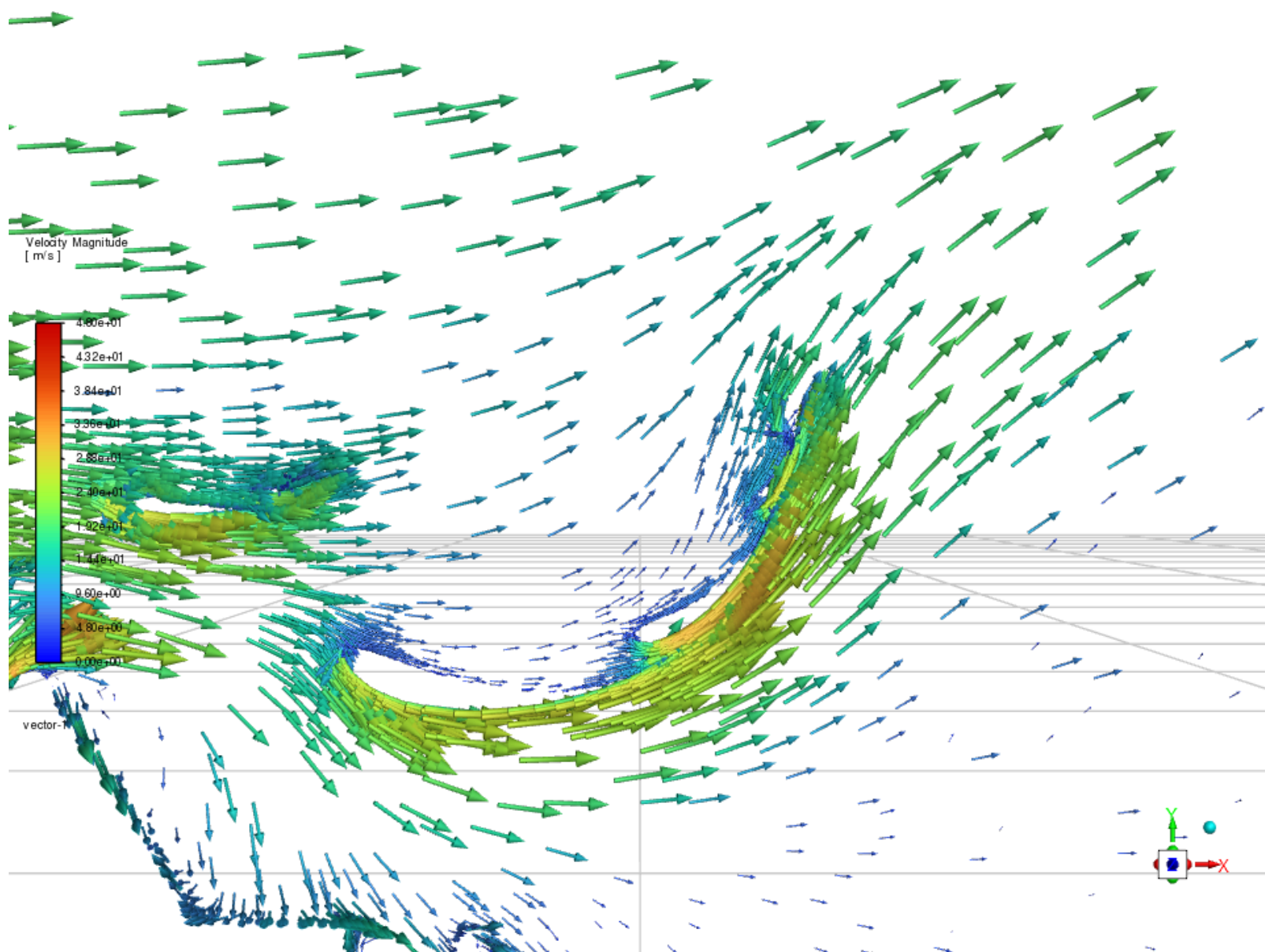
vector-3



vector-2



vector-1



vector-0

